

An Introduction to Time Series Analysis

With a Focus on Macroeconomic & Financial Applications

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Slides and Code available at `www.market-observatory.com`

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Outline

- 1 Introduction
- 2 Data Prep
- 3 Explore
- 4 Explain
- 5 Forecast

Learning Objectives

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- Understand what is “special” about time series. As such, I’ll be focusing on differences to non time series settings and techniques.
- Learn how to adapt our typical workflow for time series data
- Introduce new techniques that are not feasible within non time series settings.

Intended Audience and Seminar Structure

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- Attendees have a basic understanding of statistical inference
- Attendees have a working knowledge of standard econometric and machine learning techniques

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- Part 2: Asynchronous coding exercise; I post sample code, which you explore, complete, and execute. I will host office hours to answer questions.
- The applications will be cast within a macro and financial context, but the tools and concepts are widely applicable.

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Taxonomy of Time Series Models

- **Linear / Econometric Models**
 - AR, MA, ARIMA, ADL, ECM
- **State Space Models**
 - Kalman Filter, Structural TS, Deep State Space
- **Machine Learning Models**
 - Tree-based, Neural Networks (LSTM, N-BEATS)
- **Global / Panel Models**
 - DeepAR, hierarchical forecasting
- **Foundation Models**
 - TimeGPT, Chronos

Types of Data

Data Types:

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- Time Series $t = 1, \dots, T$
- Panel ($N \times T$)

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Modeling time series present distinct challenges from Cross Sectional or Panel data. Most important among those challenges is “temporal persistence”.

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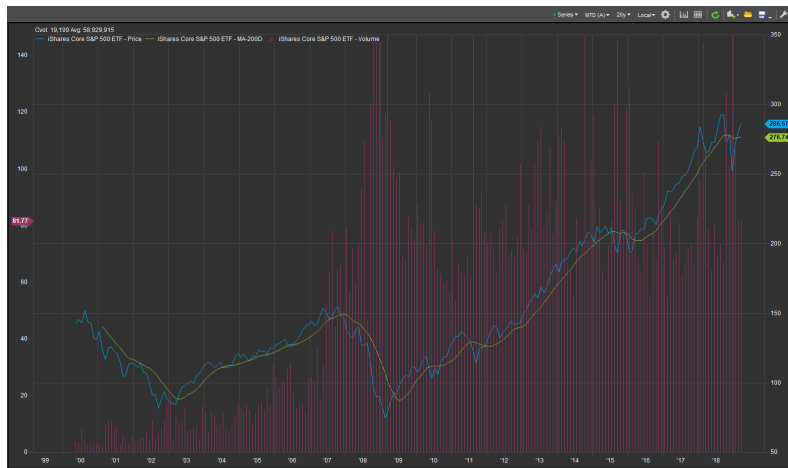
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- ② Error reduction: My error term captures relevant omitted regressors, which I don't observe or know. If these omitted regressors are persistent due to the underlying behavior, then including lagged dependent variables reduces the error.

What is so special about “time series” data?



Certain features and stylized facts appear with time series data that are not apparent in cross sectional. (e.g. momentum)

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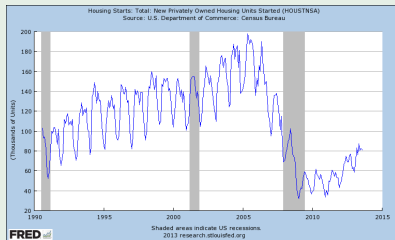
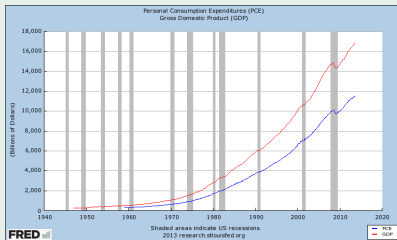
Example (Misleading Time Series Regressions)

Consider the time series models below. In each, discuss why running such a regression by OLS might be misleading.

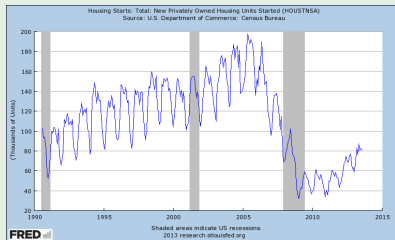
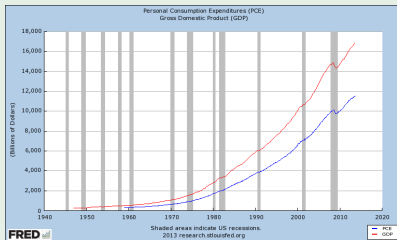
① $C_t = \beta_0 + \beta_1 Y_t + u_t$

② $HsngStarts_t = \beta_0 + \beta_1 MrtgRates_t + u_t$

Example (Misleading Time Series Regressions)

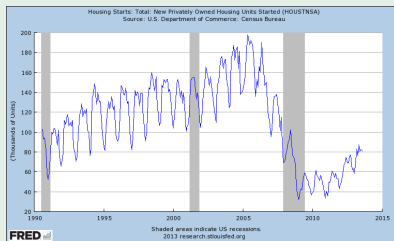
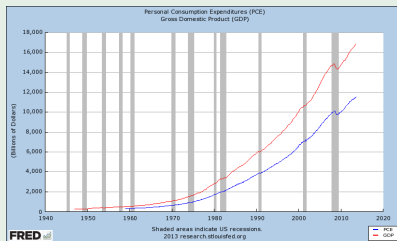


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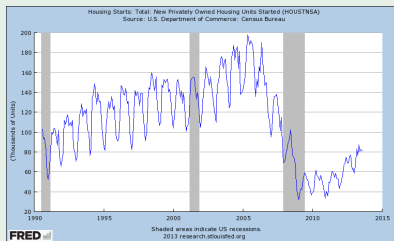
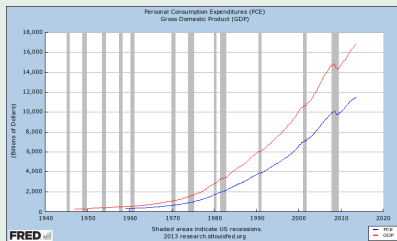
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- 2 (Right) Seasonal patterns

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We need to accommodate for these common features of time series.

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These features imply:

- Standard cross-sectional tools may fail
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Next: How does the time series workflow adapt to these challenges?

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Why Data Preparation Matters in Time Series

Time series data requires special care before modeling:

- **Ordering matters**
 - Incorrect ordering destroys temporal dependence
- **Missing data is not innocuous**
 - Dropping or imputing values can distort dynamics
- **Timing must be aligned**
 - Misaligned data introduces lookahead bias

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Poor data preparation leads to misleading inference and invalid forecasts

Sequencing Doesn't Matter for Cross Sectional Data

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Company	# Distribution Facilities	Order Fullfillment Time
ABC	10	18
DEF	12	17
GHI	13	17
JKL	15	16
MNO	19	13
PQR	20	13
STU	21	12
VWX	23	11
YZA	29	8

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Company	Year	# Distribution Facilities	Order Fullfillment Time
ABC	1990	2	26
ABC	1991	2	25
ABC	1992	4	23
ABC	1993	5	22
ABC	1994	6	22
ABC	1995	6	23
ABC	1996	7	21
ABC	1997	8	19
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The temporal dependence among observations requires that we maintain the sequencing of the time series data.

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- Convention: Oldest to Youngest

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- Convention: Oldest to Youngest
 - Caution: Not all data vendors deliver data in this way
- Pro Tip: Always sort your data before you model
- Best Practice: resort by the date/time vector after each filter/select
- For intraday day, be sure you are sorting on the entire date/time stamp, and not just the date/day/year/hr, etc..

Date Time Stamp vs Time Index

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Company	Date Time Stamp	Time Index	# Distribution Facilities	Order Fullfillment Time
ABC	1990	1	2	26
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ABC	1993	4	5	22
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- Using the Time Index is easy to implement and interpret for simple linear time trends on equally spaced data
- But if data has temporal patterns (e.g. seasonality, unequal spacing, etc..), then modeling directly on the Date Time Stamp is preferred.

Cleaning Time Series

Don't blindly drop rows

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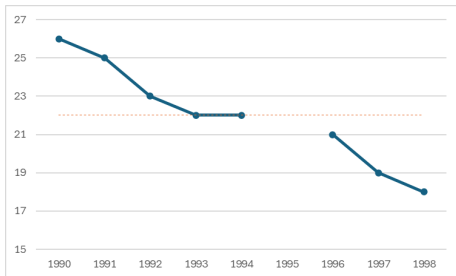
The temporal dependence of the series is disrupted when you drop rows.

Cleaning Time Series

Filling with Sample Mean can be Dangerous

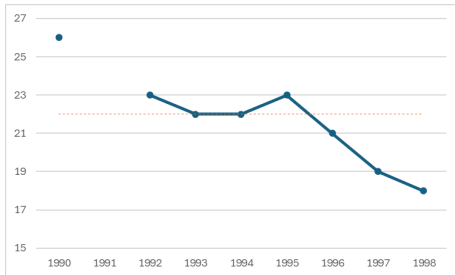
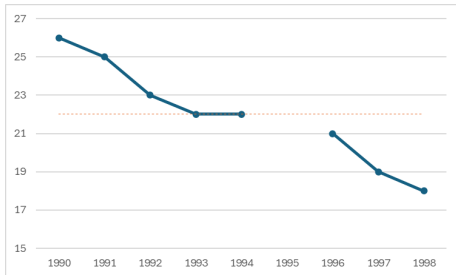
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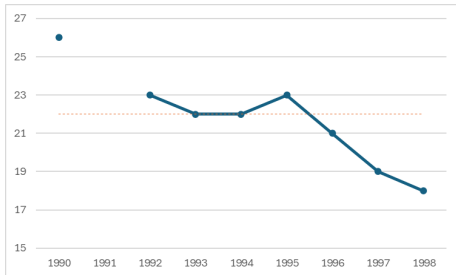
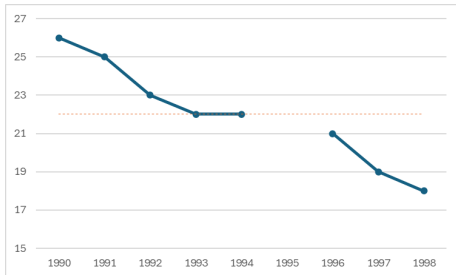
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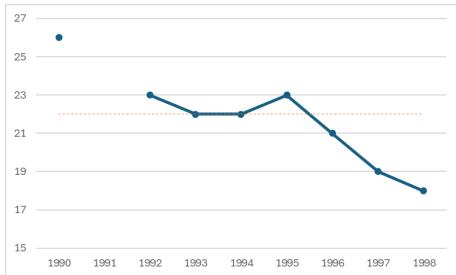
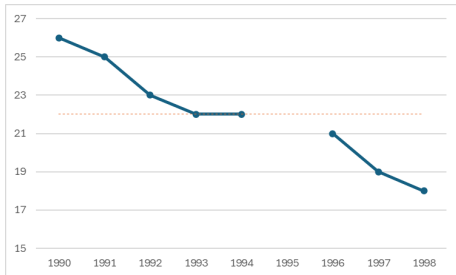
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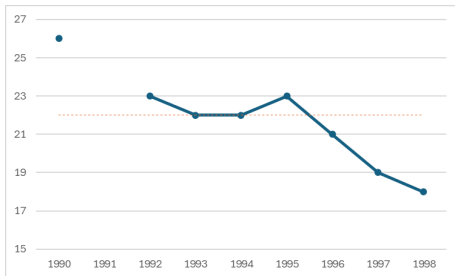
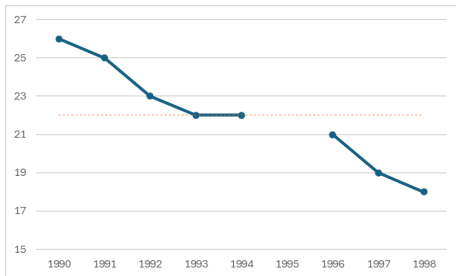


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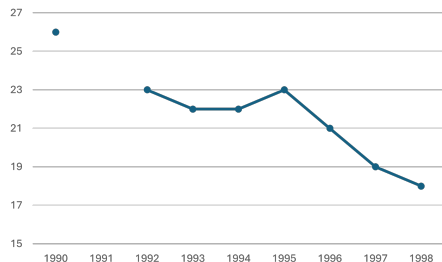
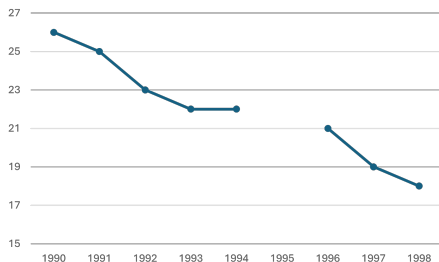
- While the mean may be a suitable proxy for the series on the left, it is not for the series on the right.
- In general, filling with the sample mean is especially dangerous on trending series

Cleaning Time Series

Interpolating is a common approach

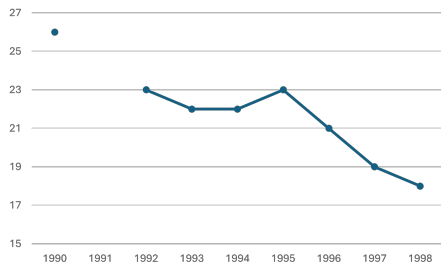
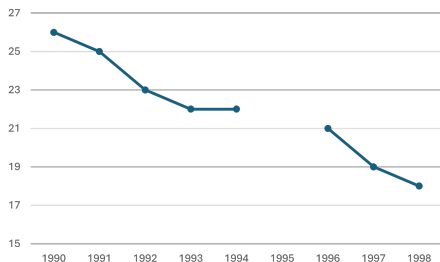
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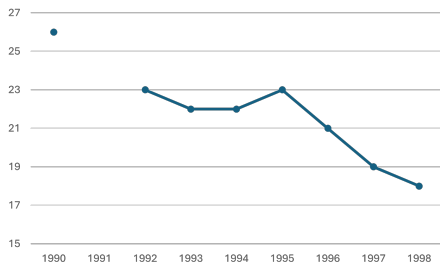
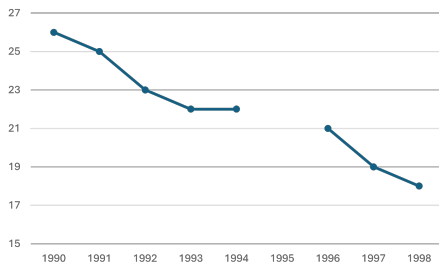
Interpolating is a common approach



- Interpolating with a local midpoint is common

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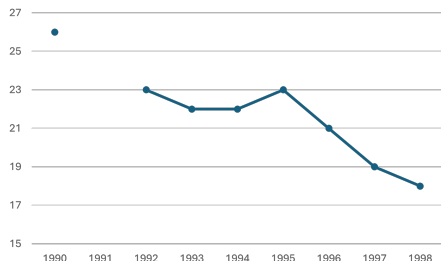
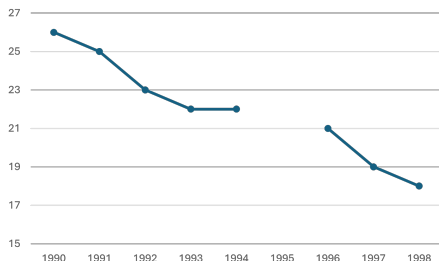
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- Using sophisticated structures (e.g. ARMA) is possible, but

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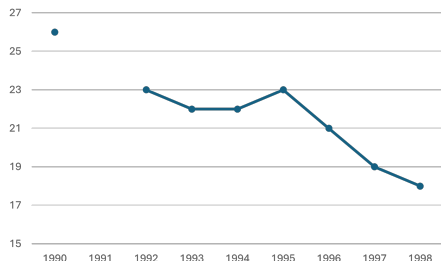
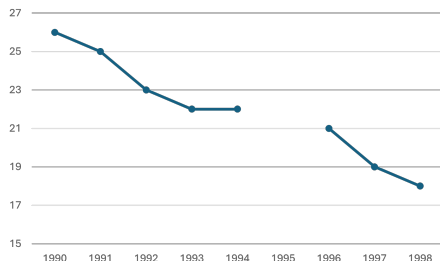
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- Interpolating with a local midpoint is common
- Using sophisticated structures (e.g. ARMA) is possible, but requires i) more parametric assumptions, and ii) more data (e.g. modeling the missing point in the right panel is more difficult)

Cleaning Time Series

Forward Filling Values

Cleaning Time Series

Forward Filling Values

Date Time Stamp	Trade Price	Last Trade Price
1/3/2025 10:01	25.25	25.25
1/3/2025 10:02	25.35	25.35
1/3/2025 10:03	26.05	26.05
1/3/2025 10:04		26.05
1/3/2025 10:05	26.10	26.10
1/3/2025 10:06	26.10	26.10
1/3/2025 10:07	26.05	26.05
1/3/2025 10:08	26.00	26.00
1/3/2025 10:09	25.95	25.95

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- Forward filling values can be sensible in some applications

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1/3/2025 10:05	26.10	26.10
1/3/2025 10:06	26.10	26.10
1/3/2025 10:07	26.05	26.05
1/3/2025 10:08	26.00	26.00
1/3/2025 10:09	25.95	25.95

- Forward filling values can be sensible in some applications
- Here, the last trade at 10:03 is the best information available at 10:04

Mixed Frequencies

Mixed Frequencies

Date	UR	GDP
Jan-20	4.2	
Feb-20	4.3	
Mar-20	9	3.2
Apr-20	12.5	
May-20	12.6	
Jun-20	10	-5
Jul-20	9	
Aug-20	9.2	
Sep-20	8	4
Oct-20	6	
Nov-20	6.5	
Dec-20	5	3

Mixed Frequencies

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Aug-20	9.2	
Sep-20	8	4
Oct-20	6	
Nov-20	6.5	
Dec-20	5	3

Q: Options for data prep to facilitate modeling?

Mixed Frequencies

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Feb-20	4.3	
Mar-20	9	3.2
Apr-20	12.5	
May-20	12.6	
Jun-20	10	-5
Jul-20	9	
Aug-20	9.2	
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Avoid Lookahead Bias

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- Simple Linear Model: $Price = \alpha + \beta Earnings + u$.
Note: We usually don't model price directly. We also need to be more precise about forecast horizon. But let us ignore for now.

Avoid Lookahead Bias

Q: What's wrong with setting up our data this way?

Date	Quarter	EPS	Stock Price
31-Mar	Q1	6.12	225
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- Solution: align release dates; e.g. if Q1 earnings were released on 24Apr, then put that EPS in the same row as the stock price on that day

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3 Explore

- AutoCorrelation Functions
- Clustering
- Dimension Reduction
- Stationarity
- Nonstationarity and Cointegration
- Decomposition, Filters, and Seasonality
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We also have a few new tools.

- Autocorrelation Function
- Partial Autocorrelation Function
- Cross Correlation Function

Autocorrelation Functions

Definition

The k^{th} autocorrelation is defined as

$$\rho_y(k) = \frac{\text{Cov}(y_t, y_{t-k})}{V(y_t)}$$

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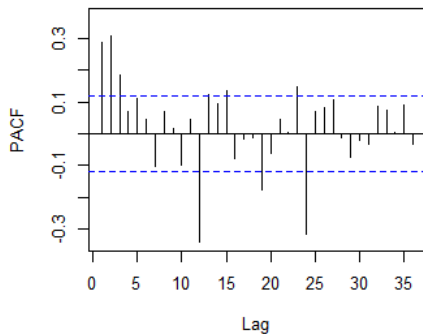
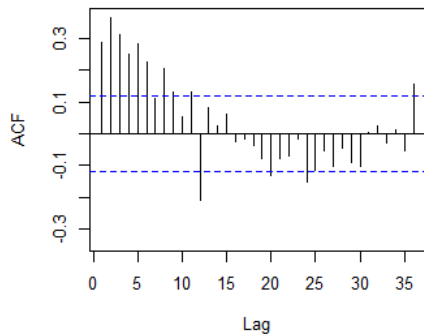
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The k^{th} partial autocorrelation is defined as the estimate of θ_k in the AR(k) autoregression. For instance

- $\text{pac}_y(1) = \hat{\theta}_1$ from $y_t = \theta_1 y_{t-1} + \varepsilon_t$
- $\text{pac}_y(2) = \hat{\theta}_2$ from $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t$

ACF and PACF



Interpreting ACF vs PACF

Q: How do we interpret $ACF(5) = 0.25$ and $PACF(5) = 0.10$?

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Since $PACF(5) < ACF(5)$, most of the dependence is transmitted through shorter lags rather than a direct effect.

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Q: What does $\rho_{r_{IBM}, r_{SP500}}(2) = 0.1$ mean?

A: Correlation between SP500 returns on “Monday” and IBM returns on “Wednesday” is 0.1; i.e. IBM responds with a two day lag.

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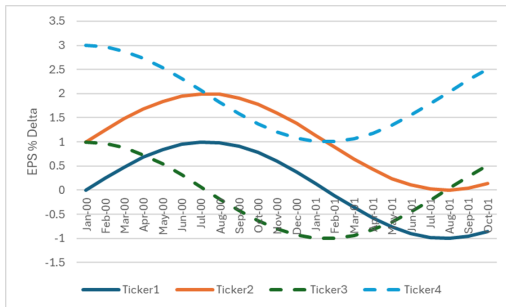
Clustering time series data takes special care

Clustering Aligned Time Series

Aligned: i) same length, ii) no leads/lags

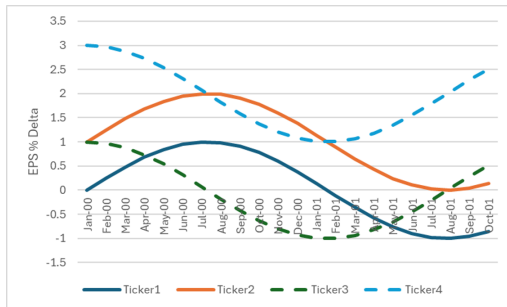
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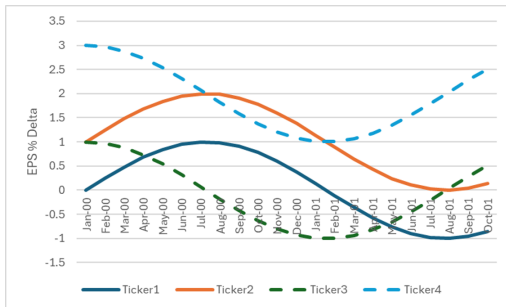
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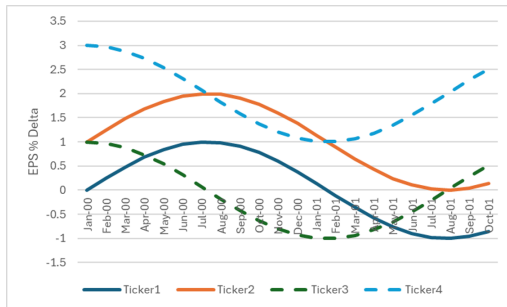
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- 1 Compute all pairwise correlations
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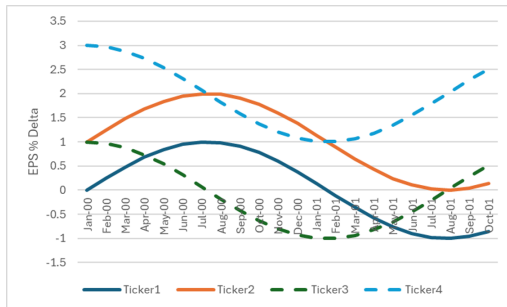
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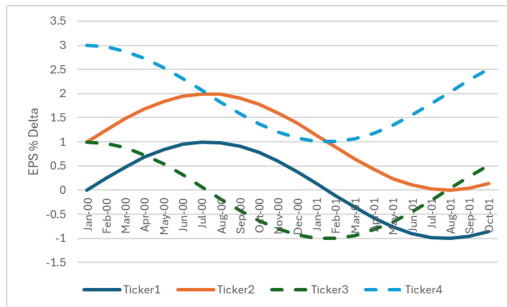
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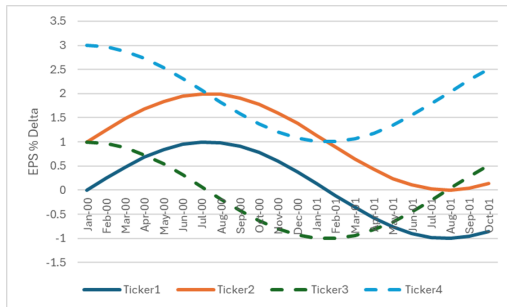
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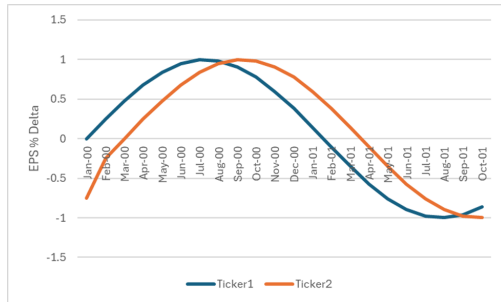
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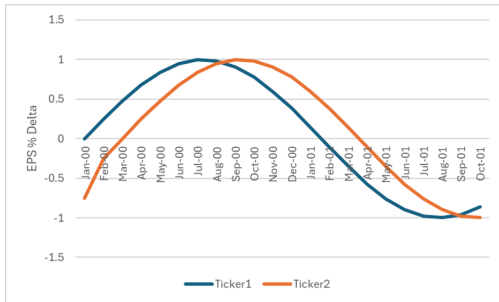
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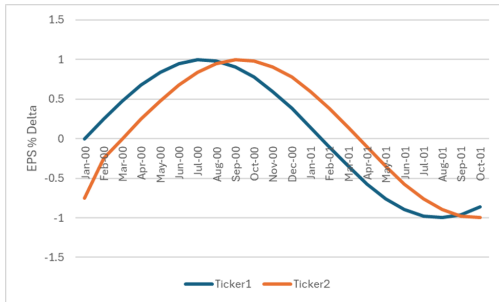


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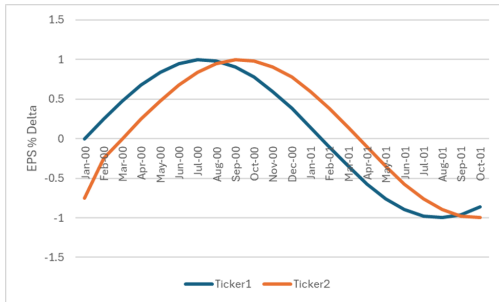
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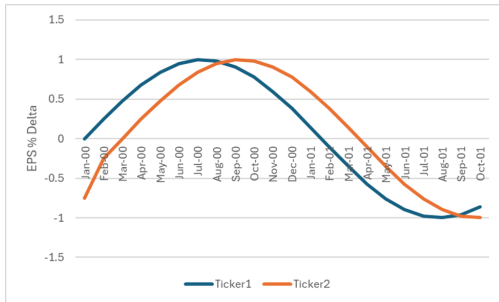
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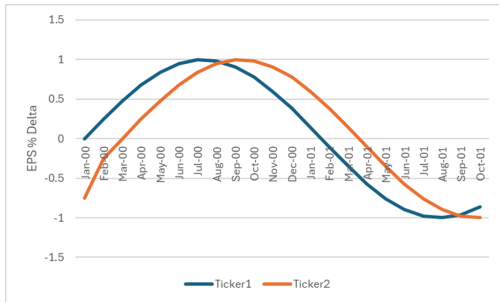
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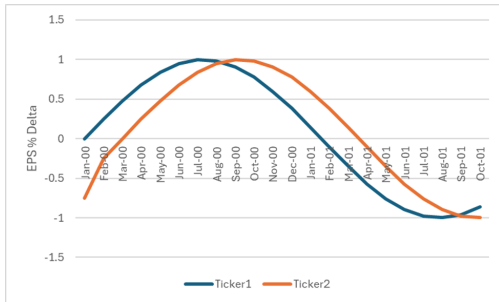
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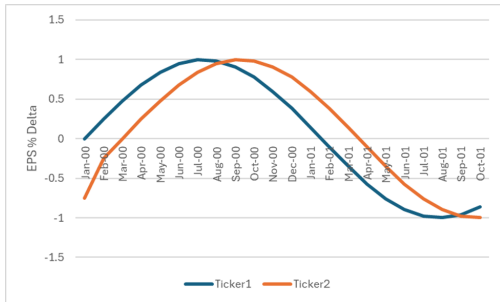
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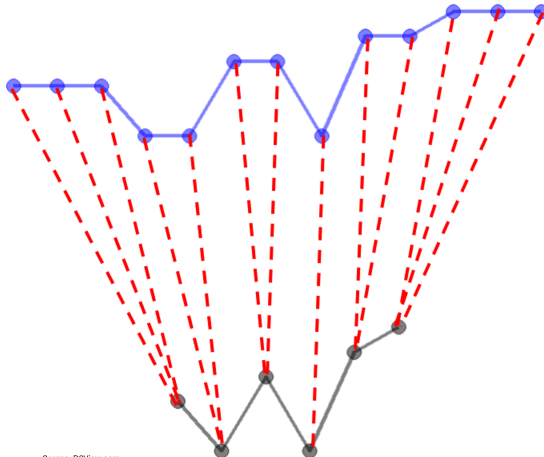
Note: Most useful when interested in shape (e.g. which assets tend to have similar trajectories)

Clustering Misaligned Time Series

Consider two assets that react similarly to macro news but at different speeds

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Source: D3View.com

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- π : alignment path between the two series
- $c(\cdot, \cdot)$: pointwise cost (e.g. squared distance)

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- π : alignment path between the two series
- $c(\cdot, \cdot)$: pointwise cost (e.g. squared distance)
- π allows matching points at different times ($s \neq t$)

Dynamic Time Warping (DTW)

- Goal: measure similarity between two time series allowing for time shifts
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- Different choices define different DTW distances:
 - Cost function
 - Warping constraints

DTW: What Are We Clustering?

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- Each entry = distance between two entire time series
- We cluster the series using this matrix
- Analogous to clustering correlations, but now using DTW distances

DTW vs Correlation Clustering

- Aligned case:
 - Similarity = correlation

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- **Same workflow:**
 - Compute pairwise similarity/distance
 - Cluster based on that matrix

Overview of How DTW Works

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- Find the “lowest-cost” path aligning the two series
- Cost reflects how much warping is required
- Output:
 - Small distance $\rightarrow$ similar shape (possibly shifted)
 - Large distance $\rightarrow$ dissimilar

DTW Handles Unequal Lengths & Flexible Alignment

- Works even if series have different lengths:

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- Final DTW distance = minimum-cost path through matrix

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When Is DTW Clustering Useful?

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- Assets reacting to the same events at different speeds
- Business cycles across countries
- Customer behavior with delayed responses
- Key benefit:
 - Groups series by **shape**, not exact timing

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1 Introduction

2 Data Prep

3 Explore

- AutoCorrelation Functions
- Clustering
- Dimension Reduction
- Stationarity
- Nonstationarity and Cointegration
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Dimension Reduction in Time Series

We often construct many related signals:

- Momentum over multiple horizons
- Macroeconomic indicators (FRED panels)
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We often construct many related signals:

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- Macroeconomic indicators (FRED panels)
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These signals are:

- Highly correlated
- Noisy
- Redundant

Goal: Reduce many signals into a few informative factors

Two Types of Dimension Reduction

Cross-Sectional Reduction (at each t):

- Many features at time t
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Temporal / Sequence Reduction:

- Long time series $\rightarrow$ compact representation
- Example: compress entire trajectory

Example: Momentum Signals

Suppose we compute momentum over multiple horizons:

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Goal:

- Combine these into a single signal z_t
- Capture common variation across horizons

PCA in Time Series: Rolling Cross-Sectional PCA

Given returns: $R_t^{LR} \in \mathbb{R}^p$

At each time t :

- Treat R_t^{LR} as a cross-section of features
- Estimate covariance matrix using a rolling window:

$$\Sigma_t$$

- Apply PCA:

$$\Sigma_t v_{1,t} = \lambda_{1,t} v_{1,t}$$

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Key idea:

- PCA is always cross-sectional
- Time series structure enters via **rolling estimation**
- Limitation: Factors are estimated independently each period (no time structure)

Autoencoders (Nonlinear PCA)

Input (same as PCA):

$$R_t^{LR} = \{R_t^{(13m)}, R_t^{(12m)}, R_t^{(11m)}\}$$

Autoencoder:

$$R_t^{LR} \xrightarrow{f_\theta} z_t \xrightarrow{g_\phi} \hat{R}_t^{LR}$$

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Output:

$$z_t \in \mathbb{R}$$

- z_t is a nonlinear factor
- Direct analog of PCA score

Autoencoder Interpretation

Optimization:

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Attempts to overcome the limitations of Rolling PCA and Autoencoders by compressing the entire time trajectory of each series

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Applications:

- Cluster assets based on full trajectory
- Identify similar patterns (trends, volatility regimes)
- Input features for downstream models

Sequence Autoencoders: Interpretation

Optimization (same as autoencoder):

$$\min_{\theta, \phi} \sum_i \left\| (r_1^{(i)}, \dots, r_T^{(i)}) - g_{\phi}(f_{\theta}(r_1^{(i)}, \dots, r_T^{(i)})) \right\|^2$$

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Limitation:

- Each series is compressed independently
- Does **not capture cross-sectional relationships**
- Good for time patterns, weak for cross-asset structure

Extension: compress entire panel jointly (panel autoencoders)

Dynamic Factor Models: Structure

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Output / Use case:

- Start with panel: 500 assets observed daily over 10 years
 $r_t \in \mathbb{R}^{500}$, $t = 1, \dots, T$
- Goal: reduce cross-sectional dimension ($500 \rightarrow 5$)
- Output: $f_t \in \mathbb{R}^5$, $t = 1, \dots, T$
- Result: **10-year $\times$ 5 factor panel** (instead of 10-year $\times$ 500 assets)

How are DFMs estimated?

- Initialize factors using PCA
- Treat model as a **state space system**
- Estimate:
 - Λ (loadings)
 - A (factor dynamics)
 - f_t (latent factors)
- Methods:
 - Kalman filter / smoother
 - EM algorithm (common in practice)

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$$DFM = PCA + \text{time dynamics} + \text{joint estimation}$$

Summary

Method	Input	Output	Structure
PCA	R_t^{LR}	z_t	Cross-section
Autoencoder	R_t^{LR}	z_t	Cross-section (nonlinear)
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All methods reduce dimension—but differ in what structure they preserve

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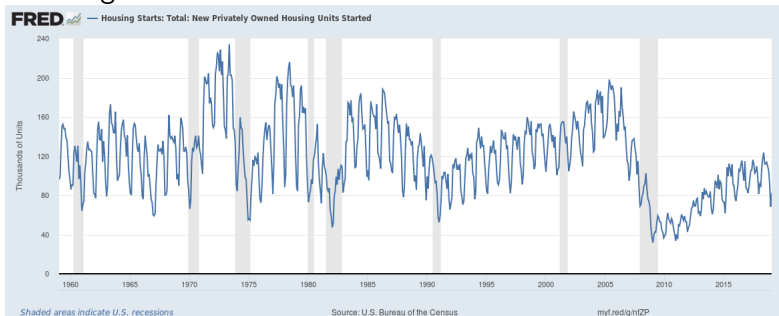
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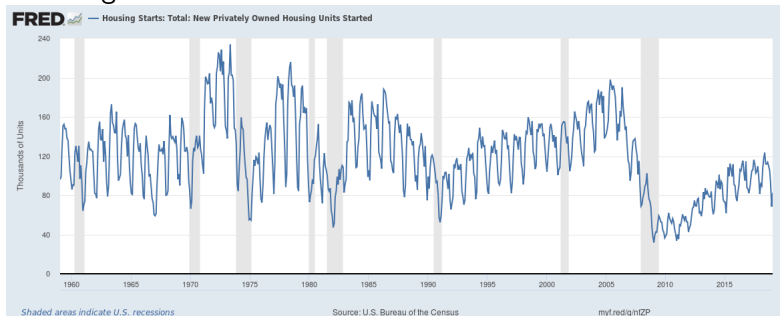
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For instance, this housing starts series exhibits clear seasonal patterns and the local mean shifts over time. These characteristics might cause trouble if not addressed.

Weak Stationarity

Definition

A process Y_t is weakly stationary if for all t it holds that

$$E[Y_t] = \mu < \infty \quad (1)$$

$$\text{Var}(Y_t) = E[(Y_t - \mu)^2] = \gamma_0 < \infty \quad (2)$$

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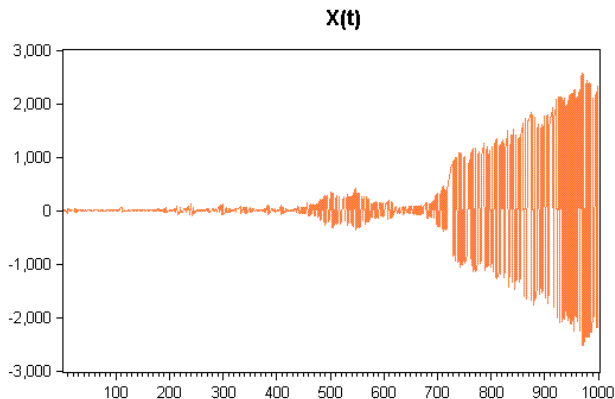
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By contrast, “Strong” or “Strict” stationarity is similar, but suggests that the entire CDF is unaffected by time shifts (as opposed to these first two moments).

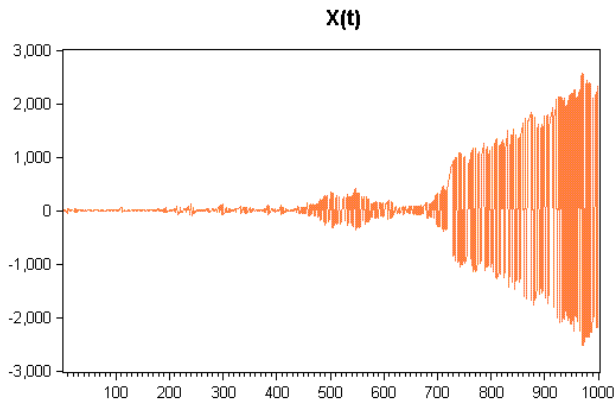
Example ((Non)Stationarity)

Q: Does the series look stationary?



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A: No. The mean is stable, but variance differs dramatically.

Example (Difference Stationary)

Consider the process: $y_t = y_{t-1} + \varepsilon_t$, where ε_t is $N(0, \sigma^2)$ iid. (This is often referred to as a Random Walk). Assume that the first observation y_0 is fixed.

- Find the mean of y_t .
- Find the variance of y_t .
- Is the series weakly stationary?

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The variance of this process diverges to infinity as time progresses. Therefore, the process is NOT stationary. We can, however, make an adjustment to engender stationarity.

Difference Stationary

We can transform a nonstationary series into a stationary series via differencing

Compute the first difference of the random walk $y_t = y_{t-1} + \varepsilon_t$:

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Second (or higher) order differences may be required to restore stationarity.

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- Stationarity is a useful property when modeling financial economic processes because it implies that our measures of mean and variance (for instance) are accurate descriptive statistics of the entire time series.
- For example, a stationary process is “mean reverting” since deviations of the mean will be neutralized in subsequent periods.
- In this section we want to explore situations when stationarity does not hold; i.e. we have Non-Stationary data

Stationarity

Let's focus on mean stationarity for now.

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Notice that $E[y_t] = c + \beta t$, which evolves non-randomly through time, and is thus NOT stationary.

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- **Stochastic Trend (Unit Root)**
 - $y_t = y_{t-1} + \varepsilon_t$
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 - Fix: **difference** the series
 - $\Rightarrow$ **Difference-Stationary (I(1))**

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The AR(1) process $y_t = \theta_1 y_{t-1} + \varepsilon_t$ can be written as:

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This last case is common in financial economic processes, and is called a unit root.

Consider the AR(1) process

$$y_t = y_{t-1} + \epsilon_t$$

Unit Roots

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For this reason, the unit root process is Non-stationary.

Random Walk

Definition (Random Walk)

Consider the non-stationary AR(1) process (i.e. $|\theta| = 1$)

$$y_t = \theta y_{t-1} + \epsilon_t = y_{t-1} + \epsilon_t$$

Then, y_t is said to follow a Random Walk.

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Then, y_t is said to follow a Random Walk.

Definition (Random Walk with Drift)

Consider the non-stationary AR(1) process (i.e. $|\theta| = 1$)

$$y_t = c + \theta y_{t-1} + \epsilon_t = c + y_{t-1} + \epsilon_t$$

where c is a constant. Then, y_t is said to follow a Random Walk with Drift.

Testing for Unit Root in AR(1)

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$$t - stat = \frac{\hat{\theta} - 1}{se(\hat{\theta})}$$

Unfortunately, that t-stat does NOT follow a t-distribution, since under the null the process is non-stationary. Dickey-Fuller (1976) tabulate critical values.

Dickey Fuller Tests for Unit Roots

Consider again the AR(1) process

$$y_t = \theta y_{t-1} + \epsilon_t$$

Without loss of generality, the Dickey Fuller tests are usually cast upon

$$\Delta y_t = (\theta - 1)y_{t-1} + \epsilon_t$$

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$$\Delta y_t = (\theta - 1)y_{t-1} + \epsilon_t$$

Here, $H_0 : \theta - 1 = 0$ implies a unit root, which is identical to the prior formulation. Under the null, y_t follows a (non-stationary) random walk. The alternative is a stationary $I(0)$ process $H_a : \theta - 1 < 0$.

Dickey Fuller Tests w/ NonZero Mean

We can recast the testing equation to include a nonzero mean for y_t

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We can recast the testing equation to include a nonzero mean for y_t

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with the null hypothesis of a unit root as $H_0 : \theta - 1 = 0$.

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Suppose the AR(1) process has a deterministic time trend

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$$y_t = c + \theta y_{t-1} + \beta t + \epsilon_t$$

If $|\theta| < 1$ and $\beta \neq 0$, y_t is nonstationary because of the time trend, not the unit root.

We can test whether y_t has a unit root (follows a random walk) vs trend stationary via $H_0 : \theta - 1 = 0$. Note: the joint hypothesis of also testing the $\beta = 0$ is often ignored.

Dickey Fuller Critical Values

Table 8.1 1% and 5% critical values for Dickey–Fuller tests (Fuller, 1976, p. 373)

Sample size	Without trend		With trend	
	1%	5%	1%	5%
$T = 25$	-3.75	-3.00	-4.38	-3.60
$T = 50$	-3.58	-2.93	-4.15	-3.50
$T = 100$	-3.51	-2.89	-4.04	-3.45
$T = 250$	-3.46	-2.88	-3.99	-3.43
$T = 500$	-3.44	-2.87	-3.98	-3.42
$T = \infty$	-3.43	-2.86	-3.96	-3.41

Figure: Source: Verbeek, A Guide to Modern Econometrics

Testing for Unit Roots in AR(2) Models

Consider an AR(2) model

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We have a unit root when $\theta_1 + \theta_2 = 1$. We can rewrite this as

$$\Delta y_{t-1} = (\theta_1 + \theta_2 - 1)y_{t-1} - \theta_2 \Delta y_{t-1} + \epsilon_t$$

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From which we can use an Augmented Dickey Fuller in order to test the null of a unit root as $H_0 : \psi = 0$.

Note that the critical values must be adjusted for this augmented test (readily available in most statistical packages).

Testing for Unit Roots in AR(p) Models

We can generalize to a test of a single root in an AR(p) process via

$$\Delta y_{t-1} = (\theta_1 + \cdots + \theta_p - 1)y_{t-1} + \delta_1 \Delta y_{t-1} + \cdots + \delta_{p-1} \Delta y_{t-p+1} + \epsilon_t$$

$$\Delta y_{t-1} = \psi y_{t-1} + \delta_1 \Delta y_{t-1} + \cdots + \delta_{p-1} \Delta y_{t-p+1} + \epsilon_t$$

with the null $H_0 : \psi = 0$ corresponding again to a unit root.

Test for Higher-order Integration

If a time series is of higher-order integration, we can find the integration order by doing a sequence of tests:

- 1 Run regression with first-difference (Δy_t) as dependent variable. If we reject the null, then y_t is $I(0)$; if we fail to reject the null, then y_t is not $I(0)$ and is at least $I(1)$, continue testing...

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- 2 ...Run regression with second-difference ($\Delta^2 y_t$) as dependent variable. If we reject the null, then Δy_t is $I(0)$ thus y_t is $I(1)$; if we fail to reject the null, then Δy_t is not $I(0)$ and is at least $I(1)$, so y_t is at least $I(2)$, continue testing ...

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- These tests are known to have low power $Pr(\text{Reject } H_0 | H_0 \text{ false})$ (i.e. too frequently misidentify stationary series as non-stationary).
- The KPSS (1992) test attempts to overcome this issue by setting H_0 : no unit (stationary) and H_a : unit root.

$$\textcircled{1} \quad Y_t = \alpha + \beta t + u_t$$

KPSS Test for Unit Roots

- ① $Y_t = \alpha + \beta t + u_t$
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- ④ Evaluate H_0 : trend stationary vs H_a : unit root.

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- ③ Construct the test statistic $KPSS = \sum_{t=1}^T \frac{S_t^2}{\hat{\sigma}^2}$, where the estimated error variance may be of the Newey-West form.
- ④ Evaluate H_0 : trend stationary vs H_a : unit root.
 - KPSS (1992) report critical values.
 - If the null hypothesis is stationary (not trend stationary), then remove the time trend from the auxiliary regression in step 1 above.

A Spurious Relationship

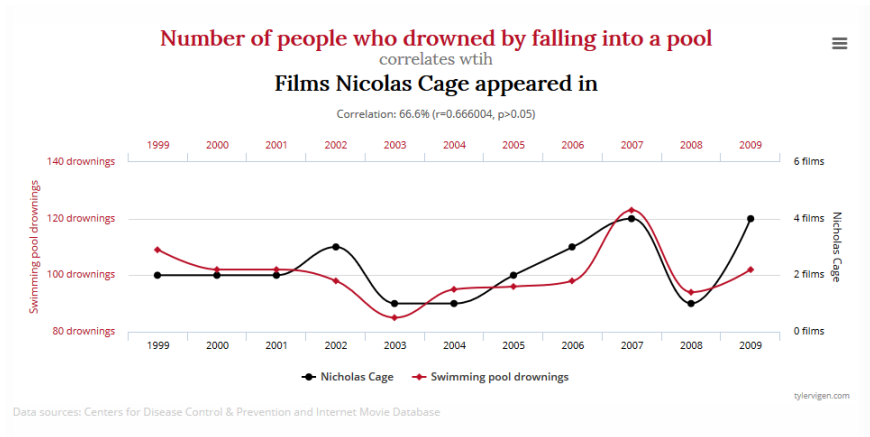


Figure: Source: tylervigen.com

A Spurious Example

A nice example from Verbeek, A Guide to Modern Econometrics:

$$Y_t = Y_{t-1} + \epsilon_{1t}, \quad \epsilon_{1t} \sim IID(0, \sigma_1^2)$$

$$X_t = X_{t-1} + \epsilon_{2t}, \quad \epsilon_{2t} \sim IID(0, \sigma_2^2)$$

where ϵ_{1t} and ϵ_{2t} are mutually independent.

A Spurious Example

A nice example from Verbeek, A Guide to Modern Econometrics:

$$Y_t = Y_{t-1} + \epsilon_{1t}, \quad \epsilon_{1t} \sim IID(0, \sigma_1^2)$$

$$X_t = X_{t-1} + \epsilon_{2t}, \quad \epsilon_{2t} \sim IID(0, \sigma_2^2)$$

where ϵ_{1t} and ϵ_{2t} are mutually independent. X and Y are non-stationary Random Walks. Nothing fundamentally links these series, but if you run

$$Y_t = \alpha + \beta X_t + \epsilon_t$$

you are likely to get a large R^2 and significant β .

A Spurious Example

Table 9.1 Spurious regression: OLS involving two independent random walks

Dependent variable: Y			
Variable	Estimate	Standard error	t -ratio
constant	3.9097	0.2462	15.881
X	-0.4435	0.0473	-9.370
$s = 3.2698$ $R^2 = 0.3072$ $\bar{R}^2 = 0.3037$ $F = 87.7987$ $dw = 0.1331$			

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Because both series are highly persistent, OLS picks up common trending behavior, producing a high R^2 even when the series are unrelated. The residuals inherit this persistence, which is why the DW statistic is low.

Cointegration Example

Consider Friedman's Permanent Income Hypothesis (PIH), which suggests that “permanent” income and consumption will grow at the same rate. Consider C^P as the long run (i.e. permanent) level of consumption and Y^P as the permanent level of income. Friedman writes this as $C_t^P = \beta Y_t^P v_t$, which we can rewrite as

$$\ln(C_t^P) = \ln(\beta) + \ln(Y_t^P) + u_t$$

If the theory is correct, then the mean of the difference in the logarithms of consumption and income will be $\ln(\beta)$.

Cointegration Example

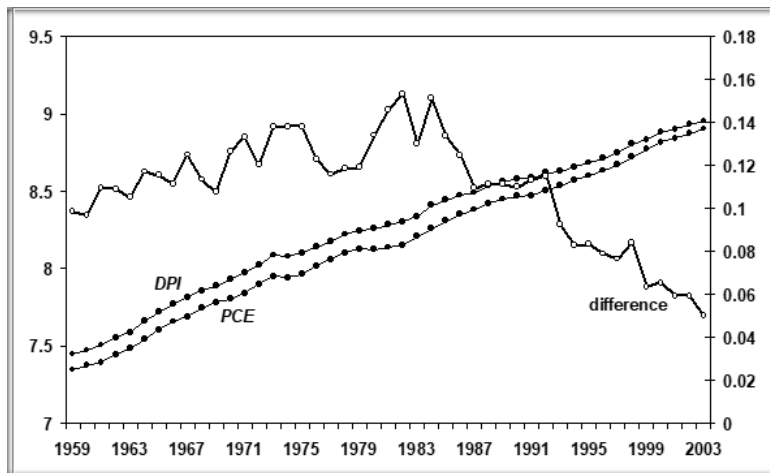


Figure: Permanent Income Hypothesis (Source: Introduction to Econometrics, Dougherty)

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- From the prior graph we see that DPI (*i.e.* Y^P) and PCE (*i.e.* C^P) each appear to be nonstationary (witness the time trends).

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- According to the theory of the PIH, these two series must drift together over time, with the short run dynamics of each canceling out.
- C^P and Y^P are said to be *Cointegrated*.

Defining Cointegration

Definition (Cointegration)

Consider $Y_t \sim I(1)$ and $X_t \sim I(1)$. Y_t and X_t are Cointegrated if there exists some value β such that $Y_t - \beta X_t \sim I(0)$.

- β : cointegration parameter
- $(1, \beta)'$: cointegrating vector
- This concept may be extended to multiple cointegrating relationships.

Warning on Inference

Suppose we estimate the following via OLS

$$Y_t = \alpha + \beta X_t + u_t$$

- Standard results tell us that $\sqrt{T}(\hat{\beta} - \beta)$ is asymptotically Gaussian. The speed at which $\hat{\beta}$ approaches β is $\sqrt{T}$ (i.e. $\hat{\beta}$ is $\sqrt{T}$ consistent for β).

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 - If, however, $\hat{\beta} = \beta$, $\hat{u}$ will be stationary with much smaller variance.
 - Since OLS minimizes the sampling variance of the residuals, it is quite good at finding an estimate close to β .

Testing for Cointegration

The key to testing for cointegration is to recognize that IF two $I(1)$ series are indeed cointegrated, the residuals from the associated linear regression are $I(0)$. Hence, we can test for a unit root. Assume series Y_t and X_t are integrated of order d , consider regression:

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$$\Delta \hat{u}_t = \beta \hat{u}_{t-1} + \delta_1 \Delta \hat{u}_{t-1} + \dots \delta_n \Delta \hat{u}_{t-n} + v_t$$

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- $\beta = 0 \Rightarrow$ Unit root $\Rightarrow \hat{u}_t$ Nonstationary $\Rightarrow Y_t$ and X_t are not cointegrated.

Cautions in using Engle Granger Test for Cointegration

- The critical value for ADF (DF) test is smaller than the usual ADF (DF) test. The reason is that by virtue of OLS's minimization of the squared residuals, the mean of the residuals is zero. Thus, the residual time series may appear more stationary than the true disturbance process.

Asymptotic critical values of the Dickey-Fuller statistic for a <u>cointegrating</u> relationship with two variables		
Regression equation contains:	5%	1%
constant, but no trend	-3.34	-3.90
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- Warning: Engle Granger is limited to single equal models. To explore cointegration in a system of equations we need other approaches, such as Johansen's test.

Cointegration in PIH

Augmented Dickey-Fuller Unit Root Test on Z		
Null Hypothesis: Z has a unit root		
Exogenous: Constant, Linear Trend		
Lag Length: 0 (Automatic based on SIC, MAXLAG=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.630141	0.7646
Test critical values1% level	-4.180911	
5% level	-3.515523	
10% level	-3.188259	

$$Z_t = \log Y_t - \log C_t$$

Figure: Source: Introduction to Econometrics, Dougherty

Given we have a univariate regression, it is sufficient to test the difference between the series via ADF. In this case we fail to reject the null, implying Z has a unit root, implying that income and consumption are NOT cointegrated. WARNING: The Engle Granger test is known to have low power against alternatives when the residuals are persistent. It is prudent to also check the autocorrelation of the series.

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1 Introduction

2 Data Prep

3 Explore

- AutoCorrelation Functions
- Clustering
- Dimension Reduction
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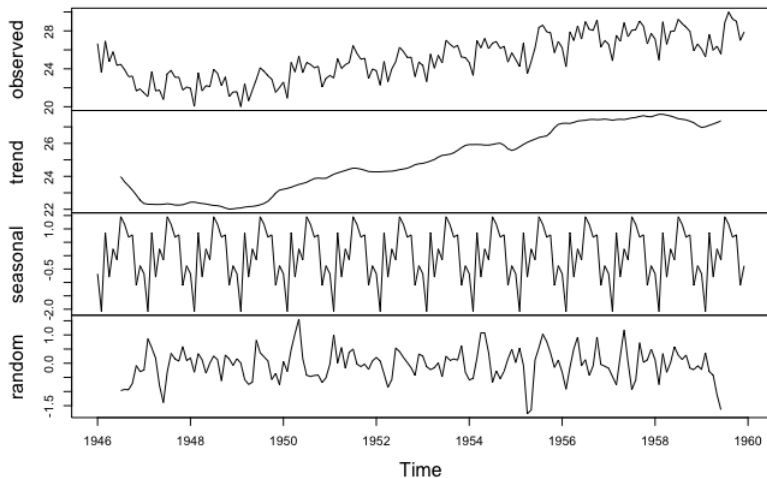
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- Interpretation:
 - A filter is a weighted average of past (and possibly future) values

Time Series Decomposition

Decomposition of additive time series



Decomposition Models

$$X_t = G_t + S_t + I_t \rightarrow \text{Additive}$$

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Note: Taking logs converts multiplicative $\rightarrow$ additive

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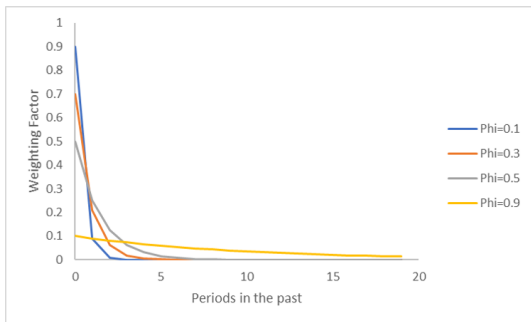
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③ Seasonally adjusted series:

$$X_t^{SA} = X_t - \bar{\bar{X}}(Qj)$$

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Diagnostics → Filtering → Model Choice

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Our diagnostics guide our modeling choices:

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Diagnostics are not just descriptive—they determine the model

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Modeling Temporal Dependence

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Key idea:

- The past helps predict the present

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We now generalize this idea with richer models

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Common Linear Time Series Models

Static : $Y_t = \beta_0 + \beta_1 X_t + u_t$

Finite Distributed Lag : $Y_t = \beta_0 + \beta_1 X_t + \beta_2 X_{t-1} + u_t$

Autoregressive Distributed Lag : $Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$

Example (A Static Time Series Model)

Suppose data on inflation π_t and the unemployment rate UR_t was gathered for 49 months, and the following simple regression was estimated.

$$\begin{aligned}\hat{\pi}_t &= 1.42 + .468 UR_t \\ &\quad (1.72) \quad (.640) \\ n &= 49, R^2 = .053, \bar{R}^2 = .033\end{aligned}$$

where standard errors are in parentheses.

- Test the hypothesis that $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 < 0$. Interpret your results.
- How do these results contradict basic macroeconomic theory? Why might that be the case?

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- We find that $t^{stat} = \frac{.468}{.64} \approx .73$. A 5% critical value for a one-tailed test with 47 degrees of freedom is approximately 1.68. This implies that we fail to reject the null.

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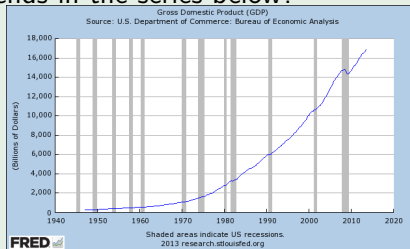
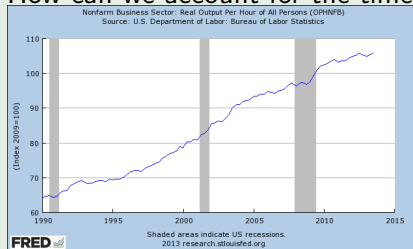
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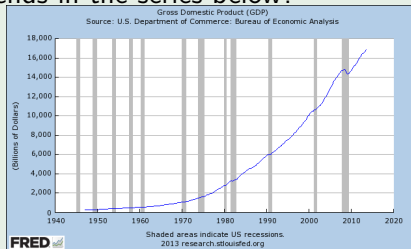
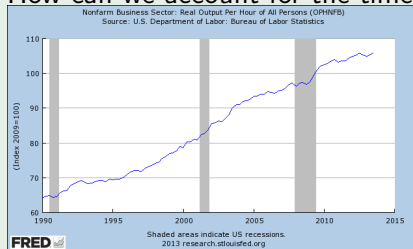
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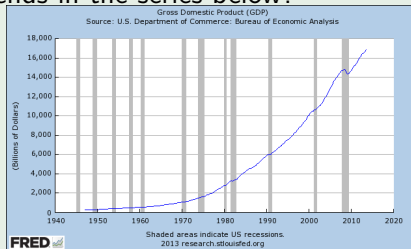
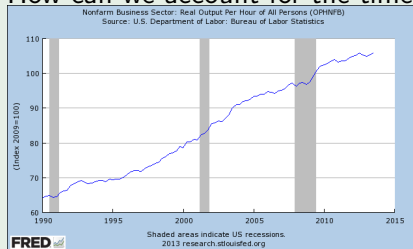
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Rearrange and take the natural log of both sides (where $\ln(e^x) = x$)

$$\begin{aligned}g &= \frac{Y_t}{Y_{t-1}} - 1 \\1 + g &= \frac{Y_t}{Y_{t-1}} \\\ln(1 + g) &= \ln\left(\frac{Y_t}{Y_{t-1}}\right) \\\ln(1 + g) &= \ln(Y_t) - \ln(Y_{t-1}) \\g &= \Delta \ln(Y_t)\end{aligned}$$

where the left hand side of the last line comes for the approximation $\ln(1 + g) \approx g$ for small g .

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suggests that $\beta = \frac{\partial \ln(Y_t)}{\partial \ln(X_t)}$. By discretizing we see $\Delta \ln(Y_t) = \beta \Delta \ln(X_t)$, and we can say “a 1% change in X corresponds to a $\beta\%$ change in Y ”. As we will see later, growth rates tend to have better properties (e.g. stationarity) than levels of macro financial time series.

Common Linear Time Series Models

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$\beta_1 + \beta_2 + \beta_3$ is the “long run multiplier”

Example (An Investment Shock)

Consider the following dynamic investment function:

$I_t = \beta_0 + \sum_{j=1}^3 \beta_j r_{t-j+1} + u_t$. Suppose the parameter estimates were $(\beta_0, \beta_1, \beta_2, \beta_3) = (5, -1, -.75, -.5)$. Moreover, assume that the interest rate (r) takes a value of 0 for all periods prior to t ; takes a value of 1 in t ; and takes a value of 0 for all periods after t .

- Provide an economic rationale for this specification.
- Compute the impact multiplier and the long run multiplier.
- Find numerical values for the level of investment in periods $t - 1$ through $t + 3$. Create an "impulse response function", which graphs I_t on the vertical axis and time periods on the horizontal axis.

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$$l_{t-1} = 5 - 1 \times 0 - .75 \times 0 - .5 \times 0 = 5$$

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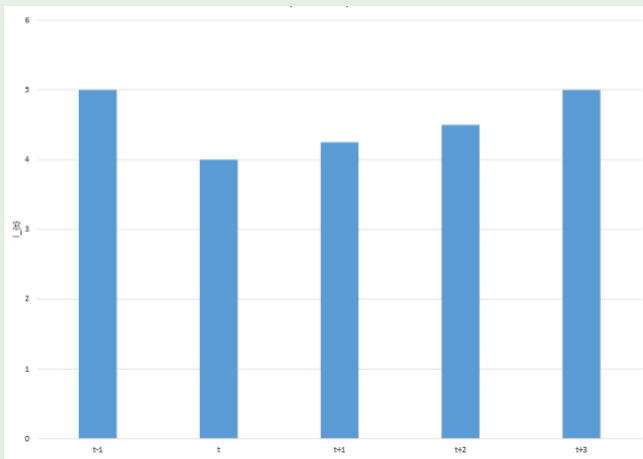
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We can interpret this Long Run Multiplier as the cumulative impact of the one unit time t shock of interest rates upon all periods of Investment.

Example (An Investment Shock)



The increase in interest rates has an immediate impact by reducing investment spending. The impact of the shock fades within 3 periods.

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Suppose you hypothesize that the value of X impacts Y for many periods.

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Solution? Include lagged values of Y instead of many lagged X 's

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$$\begin{aligned} Y_t &= \beta_1 + \beta_2 X_t + \beta_3(\beta_1 + \beta_2 X_{t-1} + \beta_3 Y_{t-2} + u_{t-1}) + u_t \\ &= (\beta_1 + \beta_3 \beta_1) + \beta_2 X_t + (\beta_3 \beta_2) X_{t-1} + \beta_3^2 Y_{t-2} + (\beta_3 u_{t-1} + u_t) \end{aligned}$$

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with continued substitution, we get something that looks like

$$\begin{aligned} Y_t = & \beta_1(1 + \beta_3 + \beta_3^2 + \cdots) + \beta_2 X_t + \beta_2 \beta_3 X_{t-1} \\ & + \beta_2 \beta_3^2 X_{t-2} + \cdots + u_t + \beta_3 u_{t-1} + \cdots \end{aligned}$$

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By estimating Equation 2 (or 3) we capture the long memory of the lagged X variables in a parsimonious manner.

ADL(p, q)

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A Subtle Concern

We just applied OLS to time series data.

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But time series data has structure. . .

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*But time series data has structure. . .
Are the usual assumptions still valid?*

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5 Forecast

Gauss Markov Assumptions for Cross Sectional Model

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Weak Persistence

Definitions vary by application, but we can think of

$$\text{Corr}(X_t, X_{t+h}) \rightarrow 0 \text{ as } h \rightarrow \infty$$

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- Independence implies $\text{Corr}(u_t, u_s) = 0 \ \forall t \neq s$.
- If violated, the disturbance term suffers from serial correlation (aka autocorrelation).

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- As opposed to contemporaneous exogeneity: $E[u_t|X_{1t}, \dots, X_{kt}] = 0$.

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- Coefficient estimators maintain conditional normality

Example (Unbiased "Slope")

Consider the simple time series regression $Y_t = \beta_1 + \beta_2 X_t + u_t$. Show that the OLS estimator of β_2 is unbiased.

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$$b_2 = \frac{\sum (X_t - \bar{X})(Y_t - \bar{Y})}{\sum (X_t - \bar{X})^2}$$

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where $a_t = \frac{X_t - \bar{X}}{\sum (X_t - \bar{X})^2}$

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Definition

An estimator is consistent if its probability limit equals the true value of the population characteristic.

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The above implies that $b_2|X \sim N(.,.)$

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When OLS Meets Time Series

Violation	What Happens	Implication
Autocorrelation	$Cov(u_t, u_{t-k}) \neq 0$	SEs wrong $\rightarrow$ invalid inference
Nonstationarity	Trends / unit roots	Spurious regression
Omitted dynamics	Missing lags	Bias ($E[u_t X_t] \neq 0$)
Endogeneity	Feedback / simultaneity	Bias + inconsistency

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Ignoring the time series aspect of your data and running a typical OLS can have major consequences. Recall the standard assumptions:

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 - But often, this is violated: $\text{Corr}(u_t, u_s) \neq 0 \ \forall t \neq s$
 - This is referred to as Serial Correlation or Autocorrelation

Consequences of Autocorrelation

Example (AR(1) Serial Correlation in a Static Model)

Consider the following model. Assume weak exogeneity ($E[u_t|X_t] = 0$), and $\bar{X} = 0$.

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

$$u_t = \alpha_1 + \alpha_2 u_{t-1} + e_t; \quad e_t \sim iid(0, 1)$$

Notice that if $\alpha_2 \neq 0$ then the serial correlation exists.

- Is the OLS estimator $\hat{\beta}_2$ unbiased?
- Is the OLS estimator $\hat{\beta}_2$ consistent?
- Is the OLS estimator $\hat{\beta}_2$ efficient?

Example (AR(1) Serial Correlation in a Static Model)

Consider the following model. Assume weak exogeneity ($E[u_t|X_t] = 0$), and $\bar{X} = 0$.

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

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$$\begin{aligned} \text{Var}(\hat{\beta}_2 | X_t) &= (\sum X_t^2)^{-2} \text{Var}(\sum X_t u_t | X_t) \\ &= (\sum X_t^2)^{-2} [\sum X_t^2 \text{Var}(u_t | X_t) + 2 \sum_{t \neq s} \text{Cov}(X_t u_t, X_s u_s | X)] \\ &= \frac{\sigma^2}{\sum X_t^2} + 2 \frac{\sigma^2}{(\sum X_t^2)^{-2}} \sum_{t \neq s} \rho_{u_t, u_s} X_t X_s \end{aligned}$$

Consequences of Serial Correlation

Good news:

- OLS coefficient estimators typically remain unbiased
- OLS coefficient estimators remain consistent
- Goodness of Fit (R^2) remains consistent

Bad News:

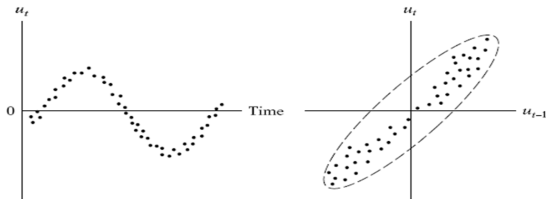
- OLS standard errors are not efficient
- Inference is no longer valid

Methods of Detection

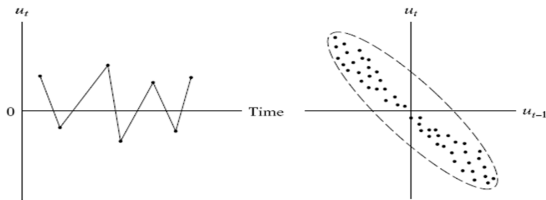
- Graphical
- Durbin Watson
- Breusch Godfrey

Example (Graphical Detection of Autocorrelation)

Given the two panels of disturbance plots nearby, which displays positive autocorrelation, and which displays negative autocorrelation?



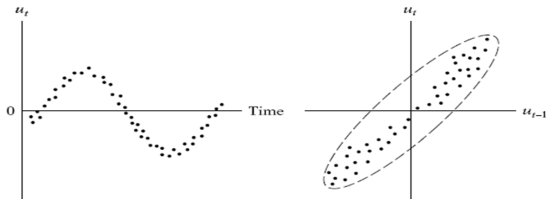
(a)



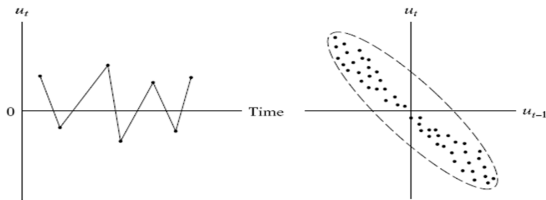
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(a)



(b)

Panel a: Positive; Panel b: Negative

$$Y_t = \beta_1 + \sum_{j=2}^K \beta_j X_{jt} + u_t \quad (8)$$

Assume all standard Gauss Markov assumptions for a time series process hold, including that X_j is strictly exogenous, except that the disturbances may be subject to AR(1) autocorrelation. Note: The regressors cannot contain lagged values of Y .

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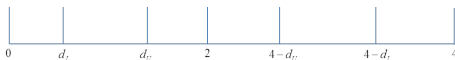
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Durbin Watson Test

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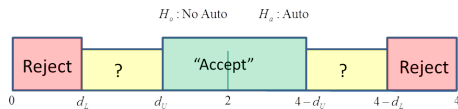
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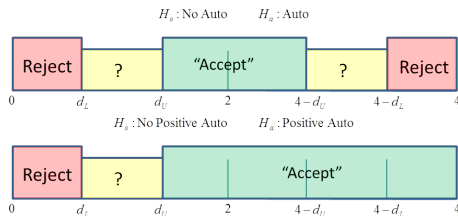


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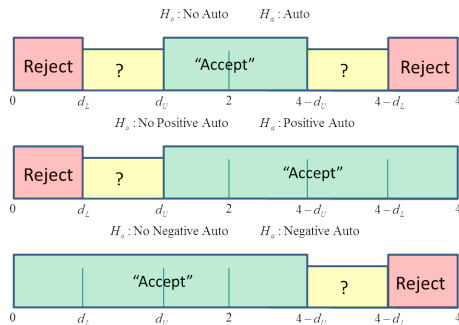
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Detection AR(1) Auto with Weakly Exogenous Regressors

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- 4 Use $t_{\hat{\rho}}$ to test $H_0 : \rho = 0$ (no auto) vs $H_a : \rho \neq 0$ (auto).

Detection AR(q) Auto. with Weakly Exogenous Regressors

Assume all standard Gauss Markov assumptions for a time series process hold, except that the disturbances may be subject to AR(q) autocorrelation, and we impose weak exogeneity.

$$Y_t = \beta_1 + \sum_{j=2}^K \beta_j X_{jt} + u_t \quad (10)$$

$$u_t = \alpha_1 u_{t-1} + \alpha_2 u_{t-2} + \cdots + \alpha_q u_{t-q} + \epsilon_t \quad (11)$$

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 - LM Test: $LM^{stat} = (T-q)R_{\hat{u}}^2$ (Breusch Godfrey)

Remedial measures

- Differencing
- Feasible GLS
- Serial Correlation Robust Standard Errors

Example (First Differencing for Serial Correlation)

Consider the simple regression $Y_t = \beta_1 + \beta_2 X_t + u_t$ where all Gauss Markov assumptions hold except that the disturbance term follows as AR(1) random walk.

- Show how first differencing can eliminate the serial correlation

Example (First Differencing for Serial Correlation)

Consider the simple regression $Y_t = \beta_1 + \beta_2 X_t + u_t$ where all Gauss Markov assumptions hold except that the disturbance term follows as AR(1) random walk.

For a random walk, the first order autocorrelation parameter is equal to 1. Therefore, $u_t = u_{t-1} + e_t$.

$$\begin{aligned} Y_t - Y_{t-1} &= \beta_2(X_t - X_{t-1}) + (u_t - u_{t-1}) \\ &= \beta_2(X_t - X_{t-1}) + (u_{t-1} + e_t - u_{t-1}) \\ &= \beta_2(X_t - X_{t-1}) + e_t \end{aligned}$$

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The $\text{Var}(\Delta Y_t | X) = \sigma_e^2$, which is homoscedastic and exhibits no serial correlation.

Feasible GLS for AR(1) Errors

$$Y_t = \beta_1 + \beta_2 X_{2t} + u_t \quad (12)$$

$$u_t = \rho u_{t-1} + e_t \quad (13)$$

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Transform (12)

$$Y_t - \rho Y_{t-1} = (1 - \rho)\beta_1 + \beta_1(X_t - \rho X_{t-1}) + e_t \quad (14)$$

$$\tilde{Y}_t = (1 - \rho)\beta_1 + \beta_2 \tilde{X}_t + e_t \quad (15)$$

where $\tilde{Y}_t = Y_t - \rho Y_{t-1}$ and $\tilde{X}_t = X_t - \rho X_{t-1}$.

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- 1 Estimate (12). Gather $\hat{u}_t$

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- ➊ Estimate (12). Gather $\hat{u}_t$
- ➋ Estimate (13). Obtain $\hat{\rho}$.

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- 1 Estimate (12). Gather $\hat{u}_t$
- 2 Estimate (13). Obtain $\hat{\rho}$.
- 3 Estimate (15) with $\hat{\rho}$.
- 4 Usual test statistics now apply.

Serial Correlation Robust Standard Errors

- 1 Estimate $Y_t = \beta_1 + \beta_2 X_{2t} + \cdots + \beta_K X_{Kt} + u_t$. Gather residuals $\hat{u}_t$ and the standard error $se^{OLS}(\hat{\beta}_2)$.

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- 2 Regress X_2 upon all the other regressors. Gather the residuals $\hat{r}_t$. Form $\hat{a}_t = \hat{r}_t \hat{u}_t$.

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- 2 Regress X_2 upon all the other regressors. Gather the residuals $\hat{r}_t$. Form $\hat{a}_t = \hat{r}_t \hat{u}_t$.
- 3 Compute $\hat{v} = \sum_{t=1}^T \hat{a}_t^2 + 2 \sum_{h=1}^g [1 - h/(g+1)] (\sum_{t=h+1}^T \hat{a}_t \hat{a}_{t-h})$, where Newey West (1987) suggest $g = \lfloor 4(T/100)^{2/9} \rfloor$.

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- 4 Compute $se(\hat{\beta}_2) = [se^{OLS}(\hat{\beta}_2)/\hat{\sigma}^2]^2 \sqrt{\hat{v}}$.

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

- Modeling Dependence (why dynamics matter)
- Linear Models (ADL, FDL)
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- Linear Model Violations (Serial Correlation)
- Rolling Regression
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Motivation for Rolling Regression

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- Current Date: Feb1980
- Data Available: Jan1970 through Jan 1980
- Estimate: $\hat{\beta}_{Jan1980}$

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- If you believe the parameter estimates are useful representations of the true DGP, then $\hat{Y} = \hat{\beta}_{Jan1980} X_{Feb1980}$

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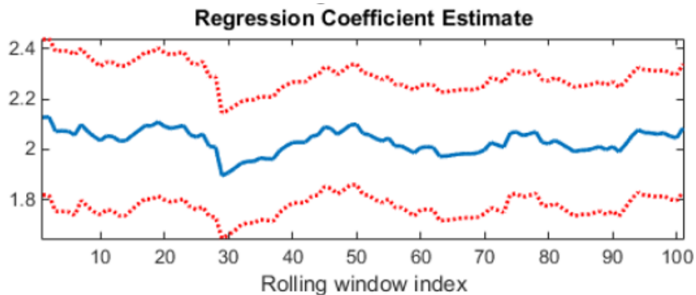
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What do you do when more data becomes available?

- If you believe the parameter estimates are useful representations of the true DGP, then $\hat{Y} = \hat{\beta}_{Jan1980} X_{Feb1980}$
- But β might be time varying.
- Typical cross validation methods ignore temporal persistence
- To accommodate we can create a time varying parameter model (beyond our purview) or sequentially update the parameter estimates.

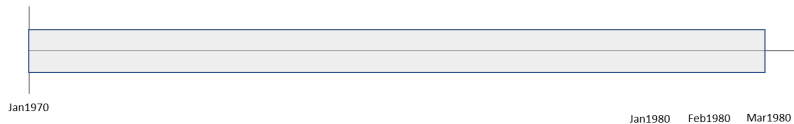
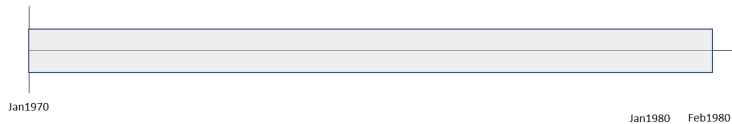
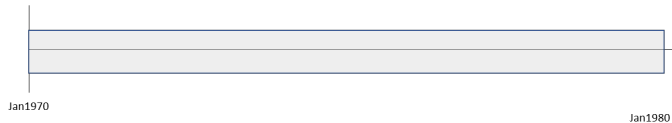
Rolling Coefficient Estimates



There are two approaches to sequentially update parameter estimates via a rolling window.

- ① Expanding Window
- ② Sliding Window

Expanding Window



Sliding Window



Rolling Estimates

Q: What are the (dis)advantages of Expanding vs Sliding Windows?

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Sliding Window

- Pro: Permits exploration of time variation in model estimates
- Con: Doesn't use all information

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Motivation ala Campbell, Lo, and MacKinlay

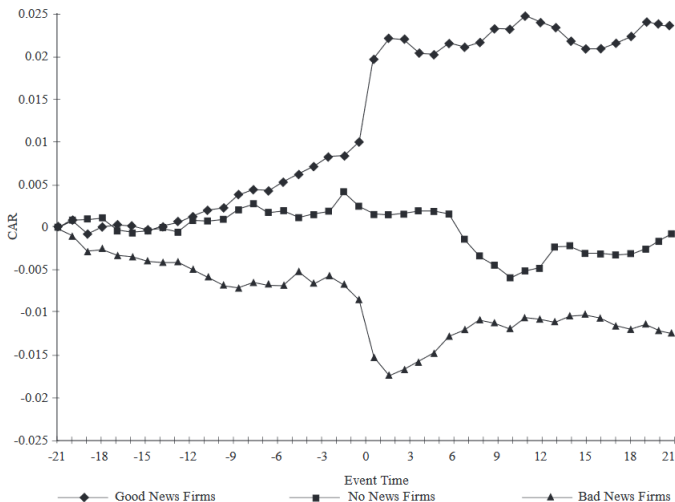


Figure 2a. Plot of cumulative abnormal return for earning announcements from event day -20 to event day 20. The abnormal return is calculated using the market model as the normal return measure.

Steps for Conducting Event Study

- 1 Define the event window
- 2 Identify universe of assets
- 3 Define baseline return model
- 4 Define the estimation window
- 5 Calculate “abnormal” returns
- 6 Conduct tests and create visualizations

Define event and estimation windows

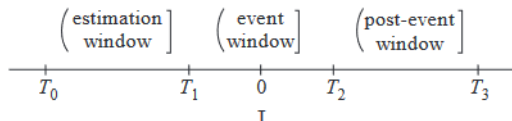


Figure 1. Time line for an event study.

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Market Model

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \eta_{i,t} \quad \eta \sim N(0, 1)$$

Abnormal Returns for Asset i

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Abnormal returns

$$AR_{i,\tau} = R_{i,\tau} - E[R_{i,\tau} | T_1] \quad \forall \tau \in \{T_1, \dots, T_2\}$$

where the $E[R_{i,\tau} | T_1]$ is the baseline model for returns of asset i using information up until the beginning of the event window T_1 . Notice that $\hat{AR} = \hat{\eta}$

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Cumulative Abnormal Returns

$$CAR_i(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} AR_{i,\tau}$$

where $T_1 < \tau_1 \leq \tau_2 \leq T_2$.

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$$J_1 = \frac{\overline{CAR}(\tau_1, \tau_2)}{[\hat{\sigma}^2(\tau_1, \tau_2)]^{.5}} \sim N(0, 1)$$

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- If J_1 is statistically significant, then we can say the “event” had an impact on the assets.
- Post estimation it is common to run a cross sectional regression on the CARs at a point in time to explore the source of variation in abnormal returns

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Modeling Long Run Relationships

Suppose there is a long run relationship between Y and X , specified as

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In the short run, the change in Y is driven by two forces:

- 1 Partial Closing of the gap between the previous LR value and the previous actual value $Y_{t-1}^* - Y_{t-1}$.
- 2 Response to the change in X during period t : $\Delta X_t = X_t - X_{t-1}$.

$$\Delta Y_t = \lambda(Y_{t-1}^* - Y_{t-1}) + \delta \Delta X_t + u_t$$

Error Correction Model

$$\begin{aligned}\Delta Y_t &= \lambda(Y_{t-1}^* - Y_{t-1}) + \delta\Delta X_t + u_t \\ &= \lambda(\gamma_1 + \gamma_2 X_{t-1} - Y_{t-1}) + \delta(X_t - X_{t-1}) + u_t\end{aligned}$$

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which is an ADL(1,1) where $\beta_1 = \gamma_1\lambda$, $\beta_2 = \delta$, $\beta_3 = 1 - \lambda$, $\beta_4 = \lambda\gamma_2 - \delta$.

Example (Error Correction)

Consider the following dynamic consumption function
 $C_t = \beta_1 + \beta_2 DPI_t + \beta_3 C_{t-1} + \beta_4 DPI_{t-1} + u_t$, with the OLS estimation below. This specification is inspired by an Error Correction model.

- Write the associated "long run" relationship.
- Write the error correction representation.
- Identify the SR MPC, the LR MPC, and the speed of adjustment. Interpret each.

Variable	Coefficient	Std Error
DPI_t	.67	.05
C_{t-1}	.5	.04
DPI_{t-1}	.33	.04
n=80, $R^2 = .35$		

Example (Error Correction)

To derive this reduced form specification estimated above, there is a long run relationship $C_t^* = \gamma_1 + \gamma_2 DPl_t + u_t$, as well as an error correction representation $\Delta C_t = \lambda(C_{t-1}^* - C_{t-1}) + \delta \Delta DPl_t + u_t$.

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- SR MPC: $b_2 = .67$
- LR MPC: $\beta_4 = \lambda\gamma_2 - \delta \Rightarrow \beta_4 = (1 - \beta_3)\gamma_2 - \beta_2$
 $\Rightarrow \gamma_2 = (.33 + .67)/.5 = 2.$

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- Speed of Adjustment: $\beta_3 = 1 - \lambda \Rightarrow \lambda = .5$: half of the discrepancy between LR consumption and actual consumption is eliminated in one period.

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Regime Modeling

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But we often interpret these data through hidden regimes:

- Expansion vs recession
- Bull vs bear markets
- High vs low volatility periods

Problem: the regime (state) is not directly observed

Latent Economic Regimes

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$$S_t = \begin{cases} 0 & \text{Expansion} \\ 1 & \text{Recession} \end{cases}$$

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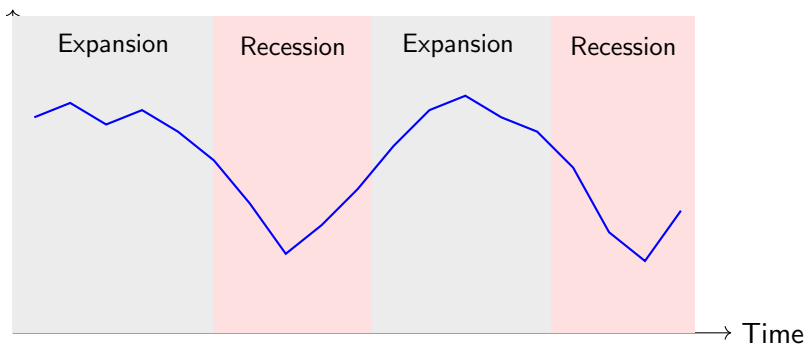
Instead, we observe:

$$Y_t = \text{economic data (GDP growth, returns, etc.)}$$

Our task is to use the observed data to infer the hidden regime.

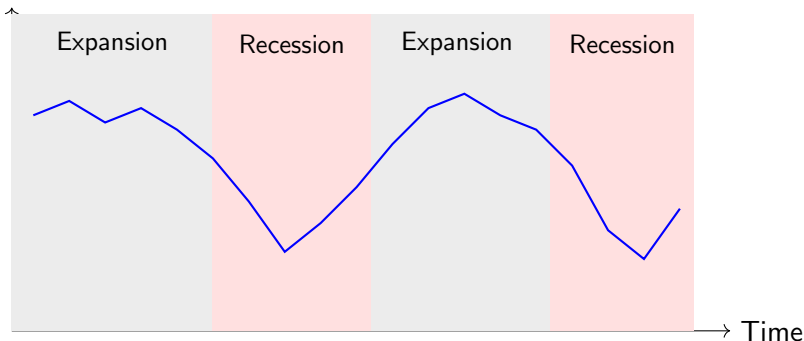
Seeing the Hidden Regimes

Growth / Returns



Seeing the Hidden Regimes

Growth / Returns



- We observe the blue time series
- We do *not* directly observe the shaded regimes
- State space models infer those hidden states from the data

State Space Representation

State evolution:

$$P(S_t \mid S_{t-1})$$

Observation equation:

$$P(Y_t \mid S_t)$$

State evolution:

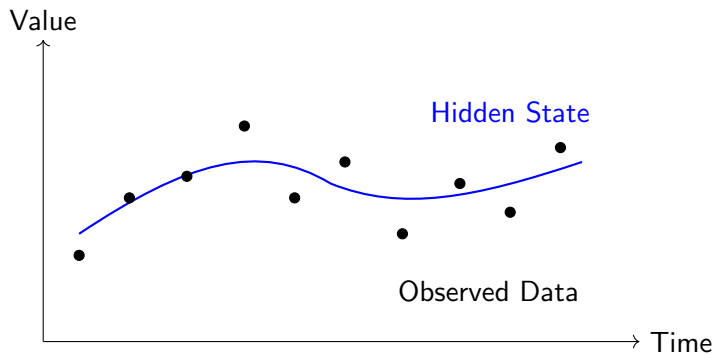
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Observation equation:

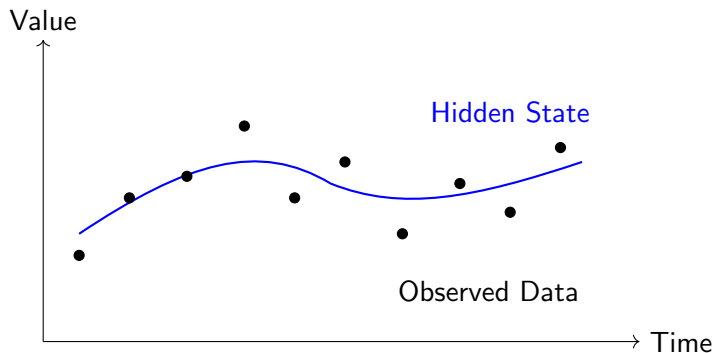
$$P(Y_t \mid S_t)$$

- The state S_t is hidden
- Observations Y_t provide noisy signals about S_t
- We update beliefs $P(S_t \mid Y_1, \dots, Y_t)$ over time

Hidden State vs Observed Data

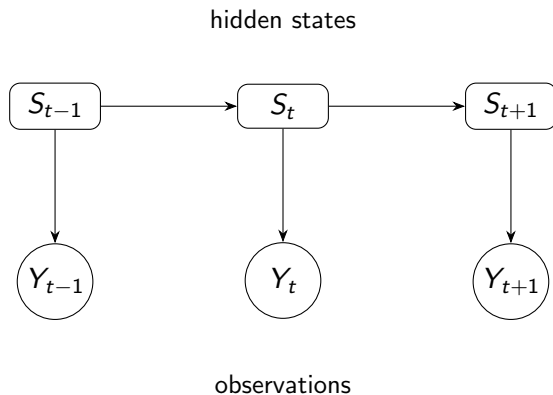


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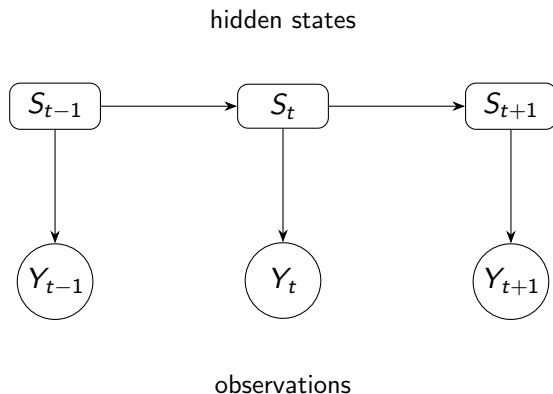


- The blue line is the latent process
- The black points are noisy observations
- Filtering and smoothing recover the hidden path

State Space Diagram



State Space Diagram



Hidden states evolve over time and generate observed data

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Collect these into a transition matrix:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix}$$

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Observations depend on the current state:

$$Y_t \mid S_t = j \sim F(\theta_j)$$

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In practice, HMMs are useful for regime detection and classification.

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Interpretation:

- In one regime, mean growth may be high
- In another, mean growth may be low or negative
- Volatility may also differ across regimes

What Changes Across Regimes?

In a Markov switching model, parameters can depend on the state:

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Examples:

- expansion vs recession in macroeconomics
- bull vs bear market in asset returns
- low vs high volatility periods in finance

Interpreting Regime Probabilities

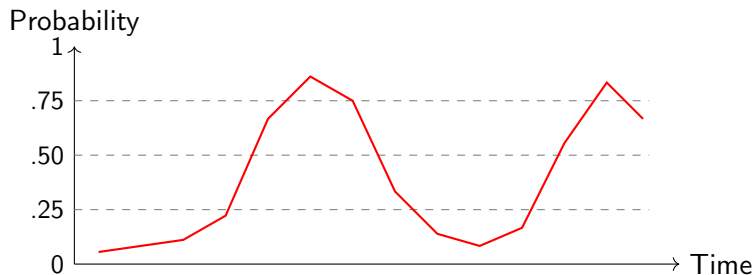
At each time t , we estimate:

$$P(S_t = \text{recession} \mid Y_1, \dots, Y_t)$$

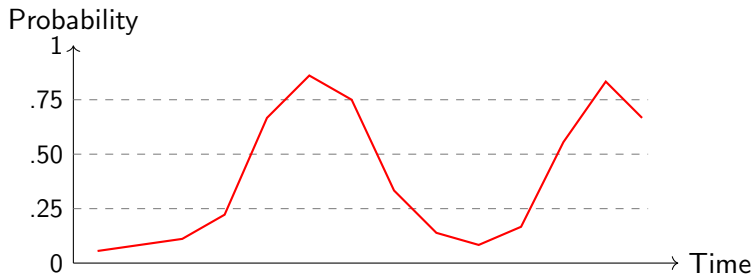
These probabilities evolve as new data arrive.

- Near 0: expansion is more likely
- Near 1: recession is more likely
- Intermediate values: uncertainty about the regime

Example Output: Probability of Recession



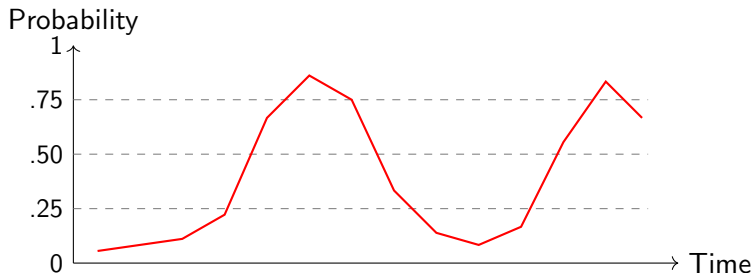
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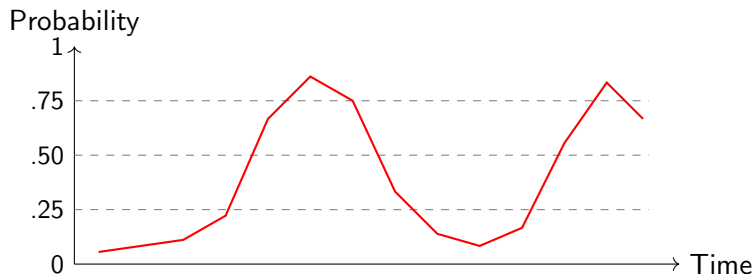


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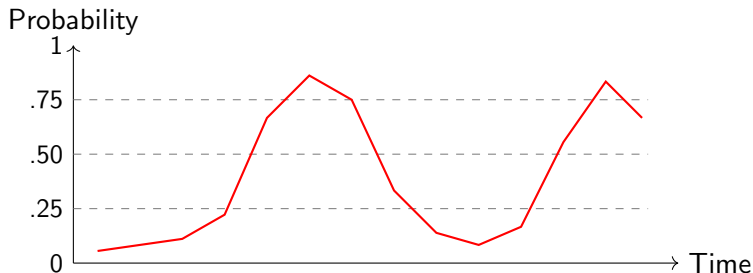


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- Optional: convert probabilities to discrete regimes using a threshold (e.g. if probability > 0.5 then label as Recession)

Deep State Space Models

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Deep state space models replace those equations with neural networks:

$$z_t = f_{\theta}(z_{t-1}, u_t, \epsilon_t)$$

$$Y_t = g_{\theta}(z_t, \eta_t)$$

where:

- z_t is a latent continuous state
- u_t may include inputs or covariates
- f_{θ}, g_{θ} are flexible nonlinear functions

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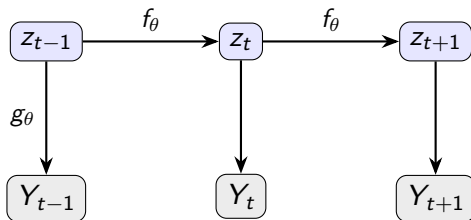
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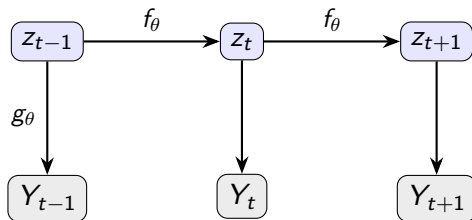
Limitations:

- Harder to interpret
- More difficult to estimate
- Usually require more data

Deep State Space Network Diagram



Deep State Space Network Diagram



- Hidden states still evolve over time
- Neural nets learn the transition f and observation g mappings

Classical vs Deep State Space

Feature	Classical	Deep
Latent state	Yes	Yes
Linear-Gaussian structure	Often	Not required
Interpretability	Higher	Lower
Nonlinearity	Limited	Stronger
Data requirements	Lower	Higher

Both aim to recover hidden structure; they differ in flexibility and interpretability

Key Takeaway

- State space models separate:
 - hidden structure (state)
 - observed data
- They provide a framework for:
 - filtering (real-time estimation)
 - smoothing (historical inference)
 - forecasting (infer current state from data, then propagate forward via transition dynamics)
- In macro and finance, they are useful for:
 - regime detection
 - trend-cycle decomposition

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

- Modeling Dependence (why dynamics matter)
- Linear Models (ADL, FDL)
- Linear Model Assumptions
- Linear Model Violations (Serial Correlation)
- Rolling Regression
- Event Study
- Long Run Structure (ECM)
- State Space
- Paradigm Comparison

5 Forecast

Comparing Time Series Modeling Approaches

Approach	Strengths	Limitations
Linear Models	Interpretable Economic structure Works with small data	Restrictive Linear dynamics
State Space Models	Flexible Captures latent components Unifying framework	More complex Requires specification
Machine Learning	Strong predictive performance Captures nonlinearities	Less interpretable Data intensive
Global / Foundation Models	Leverages many series Pretrained / scalable	Less transparent Harder to control

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5 Forecast

- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
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Outline

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- Forecasting Framework
 - What is Forecasting?
 - Forecasting Approaches
 - Forecasting Workflow
 - Challenges with Forecasting
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How Evaluate?

- Summary metrics, e.g. MSFE (Mean squared forecast error)
- Forecast error plots with time and covariates
- Forecast error densities

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- Data may have so much noise that you can't verify how wrong you are
- If you need to forecast an “X” to predict your “Y”, your errors will compound; e.g.

All Forecasts Are Wrong

- Your model may be wrong
 - ... in sample
 - ... out of sample
- Your parameter estimates may be wrong
- Data may have so much noise that you can't verify how wrong you are
- If you need to forecast an "X" to predict your "Y", your errors will compound; e.g. If I need to forecast the market's return tomorrow in order to predict my stock's return tomorrow, the errors in my market forecast will amplify my stock return forecast errors

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- Advice: never settle for a single point forecast; provide a range of values

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

5 Forecast

- Forecasting Framework
- Setup & Evaluation
 - Anatomy of a Forecast
 - Cross Validation
 - Forecast Evaluation
 - Forecast Horizons
 - Basic Forecasting Models
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning

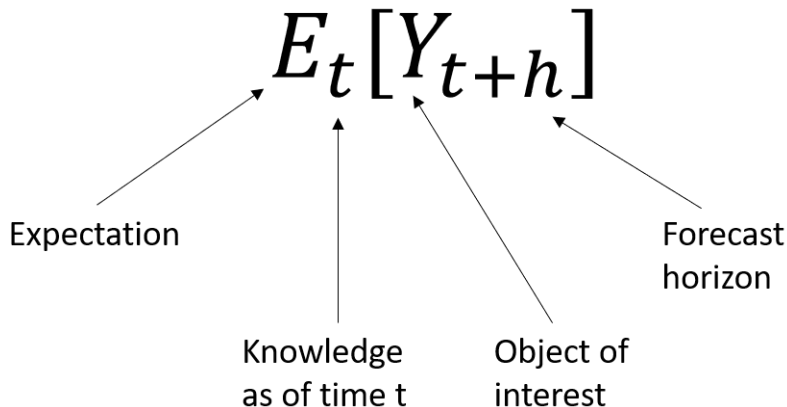
Anatomy of a Forecast

Q: Can you explain the components of this forecast?

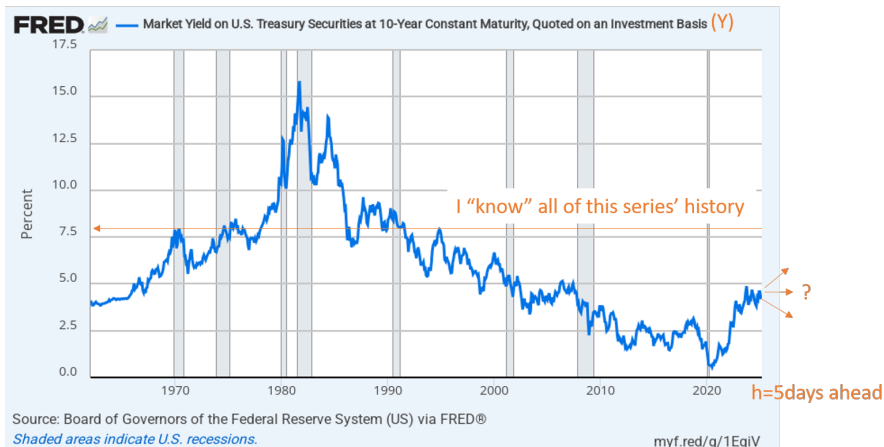
$$E_t[Y_{t+h}]$$

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5day ahead US Treasury yield forecast



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 - aka In-Sample, Estimation Period

Train / Test Split

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 - aka Out of Sample

Typical Train / Test Split

Q: Consider the data below. How do we determine the train/test split?

Company ID	Q1 Revenue
ABC	#
DEF	#
GHI	#
JKL	#
MNO	#
PQR	#
STU	#
VWX	#
YZA	#

Typical Train / Test Split

A: Random assignment of each “row”.

Train		Company ID	Q1 Revenue	Test
		ABC	#	←
→		DEF	#	
→		GHI	#	
		JKL	#	←
→		MNO	#	
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→		PQR	#	
		STU	#	←
→		VWX	#	
→		YZA	#	

The fraction of the sample assigned to train is usually higher than to split; e.g. 80% to 20%. This is a nuisance parameter that you should stress test.

Time Series Train / Test Split

Q: Why isn't the typical approach (outlined on prior slide) appropriate for our usual forecasting applications?

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Train		Time	Revenue for ABC	Test
		Q1`15	#	←
→		Q2`15	#	
→		Q3`15	#	
		Q4`16	#	←
→		Q1`17	#	
→		Q2`17	#	
		Q3`17	#	←
→		Q4`17	#	
→		Q1`17	#	

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→		Q1`17	#	
→		Q2`17	#	
		Q3`17	#	←
→		Q4`17	#	
→		Q1`17	#	

A: Randomization disrupts the temporal dependence.

Proper Splitting with Time Series data

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		Q3`17	#	←
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→		Q1`18	#	

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→		Q1`17	#	
→		Q2`17	#	
		Q3`17	#	←
→		Q4`17	#	
→		Q1`18	#	

Again, the size of the split needs to be stress tested.

Dangers with Test/Train Splitting

Q: What's the danger in this split? (Notice the different dates)

Train	Time	Revenue for ABC	Test
→	Q3`18	#	
→	Q4`18	#	
→	Q1`19	#	
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	Q2`20	#	←
	Q3`20	#	←

A: COVID (2020) likely disrupted the historical patterns. The training sample is not representative of the testing sample.

Cross Validation in Time Series

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A: Step 1: Separate the data into k-folds

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Fold 1	Q3`18	#
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	Q2`20	#
	Q3`20	#

Note: we should stress test fold size.

Cross Validation in Time Series

Cross Validation in Time Series

Step 2:

Cross Validation in Time Series

Step 2: Train on Fold 1 and Test on Fold 2

Cross Validation in Time Series

Step 2: Train on Fold 1 and Test on Fold 2

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	Q3`19	#	←
	Q4`19	#	←
Fold 3	Q1`20	#	
	Q2`20	#	
	Q3`20	#	

Note: there are many variants, which alter size and shape of testing folds, etc...

Cross Validation in Time Series

Cross Validation in Time Series

Step 3:

Cross Validation in Time Series

Step 3: Train on Folds 1 and 2 combined, Test on Fold 3

Cross Validation in Time Series

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Repeat for all of your k folds.

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- Compute $\sqrt{\frac{1}{F} \sum_{f=1}^F (\hat{\beta}_f - \bar{\hat{\beta}})^2}$ as the standard deviation of the estimates
- Other sample metrics are computed easily.

Forecast Evaluation Measures

$$Y_{t+h|t} = \beta_1 + \beta_2 X_t + u_{t+h}$$

$$\hat{Y}_{t+h|t} = \hat{\beta}_1 + \hat{\beta}_2 X_t$$

$$\hat{u}_{t+h|t} = Y_{t+h} - \hat{Y}_{t+h|t}$$

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- $ME = \frac{1}{T} \sum_{t=1}^T \hat{u}_{t+h|t}$
- $MSE = \frac{1}{T} \sum_{t=1}^T \hat{u}_{t+h|t}^2$
- $RMSE = \sqrt{MSE}$
- $MAE = \frac{1}{T} \sum_{t=1}^T |\hat{u}_{t+h|t}|$
- $MAPE = \frac{1}{T} \sum_{t=1}^T \left| \frac{\hat{u}_{t+h|t}}{Y_{t+h}} \right|$

Forecast Evaluation Test

Diebold-Mariano

Consider the errors from forecasting strategies: $\hat{u}_{t+h|t}^{(A)}$ and $\hat{u}_{t+h|t}^{(B)}$.

Impose a loss function: $L(\hat{u}_{t+h|t}^{(i)}) = (\hat{u}_{t+h|t}^{(i)})^2$

Define $\bar{d} = \frac{1}{T} \sum_{t=1}^T [L_t^{(A)} - L_t^{(B)}]$

$$H_0 : E[d] = 0$$

$$H_a : E[d] \neq 0$$

$$DM = \frac{\bar{d}}{V(\bar{d})^{1/2}} \sim N(0, 1)$$

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- Repeat the above diagnostics for squared errors to detect changes in the variability of the forecast accuracy
- Q: If you see a pattern in the error squared but not the errors, what does that signal?
- A: The model forecasts may not be biased, but the uncertainty of those forecasts vary through time or across Y or X

How Forecast?

Q: You've found the model with the best error properties. How forecast out of sample?

How Forecast?

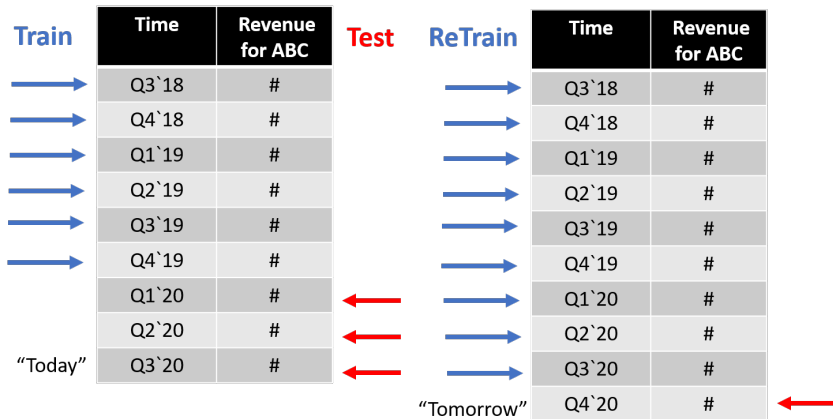
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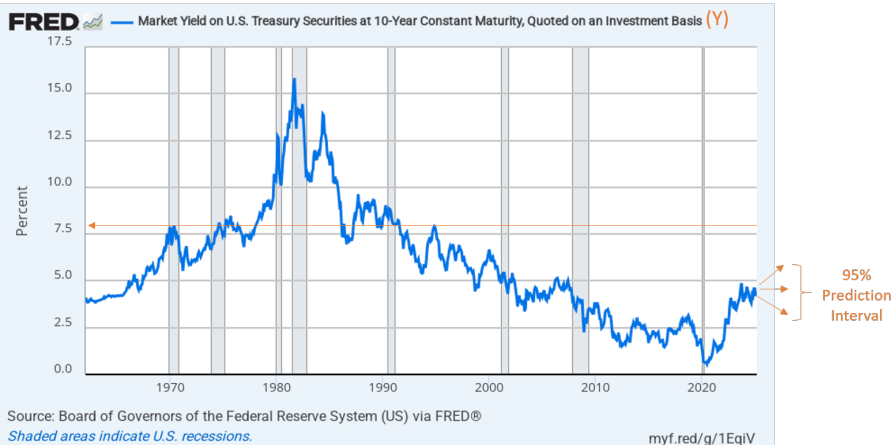
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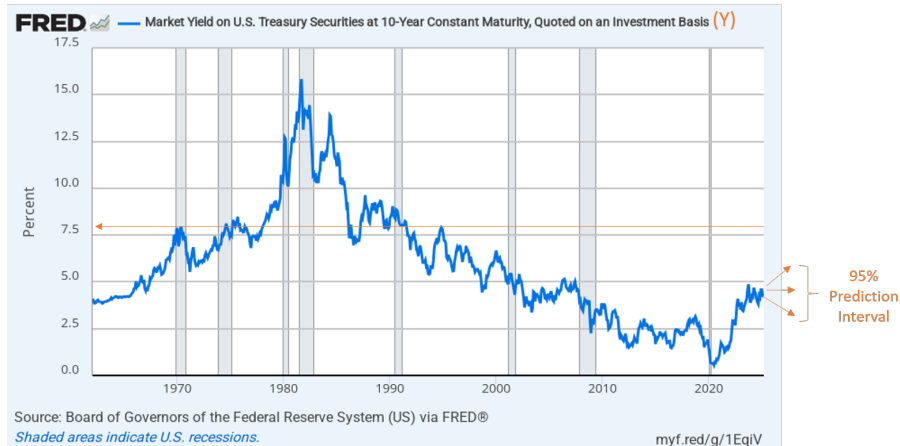
Prediction Intervals

Q: You've forecasted the 10yr yield at 4.75%. What does a 95% prediction interval of .75 mean?



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A: There is a 95% chance that the interval (4-5.5) will cover the future observation.

Forecast vs Confidence Intervals

Q: Which is wider?

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Q: Which is wider?

A: The confidence interval is:

$$\hat{Y} \pm t_{\alpha/2, n-2} \cdot se_{errors} \cdot \sqrt{\left(\frac{(x - \bar{X})^2}{SS_X} + \frac{1}{n} \right)}$$

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- Forecast Intervals are wider (notice the extra “1”). Confidence intervals reflects uncertainty about the mean prediction. Forecast intervals reflects uncertainty about the mean and the individual observation for which you are forecasting.

One Step Ahead Forecasting

One Step Ahead could literally mean tomorrow...

One Step Ahead Forecasting

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	Time	Revenue for ABC
	1Jan	#
	2Jan	#
	3Jan	#
	4Jan	#
	5Jan	#
	6Jan	#
	7Jan	#
	8Jan	#
"Today"	9Jan	#
"Tomorrow"	10Jan	#

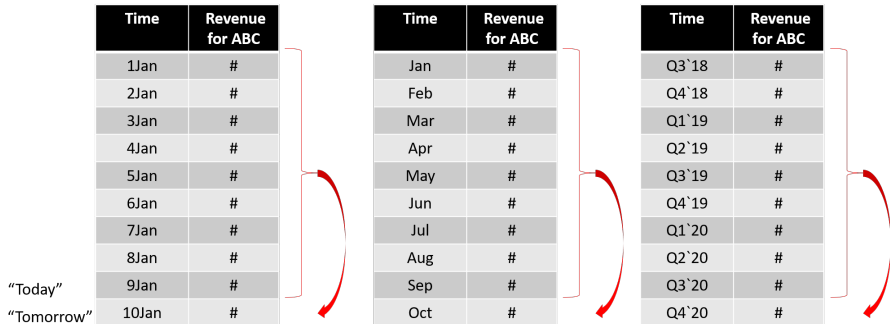
A red bracket on the right side of the table spans from the row for 9Jan to the row for 10Jan. A red arrow curves from the right side of the 9Jan row down to the 10Jan row, illustrating a one-step forecast.

One Step Ahead Forecasting

...or could mean the next period

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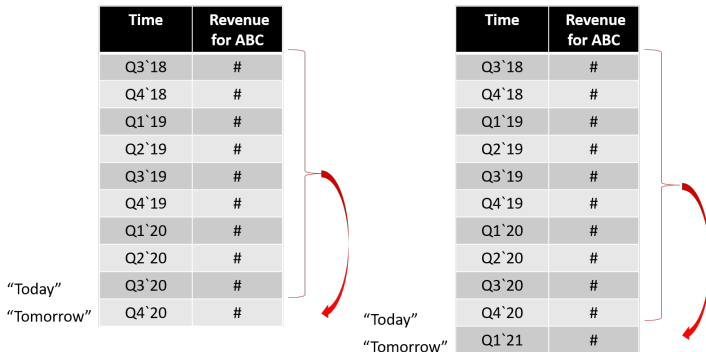
Rolling Forecasts

Q: What is a one step ahead rolling forecast?

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A: Each period we re-estimate the model to produce a one step ahead forecast.



Note: The training windows can i) retain a fixed length and slide forward or ii) grow (as shown here).

Rolling via Re-Forecasting or Re-Estimating

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which re-estimates the parameters on the new training period and then feeds in the new data for NGDP.

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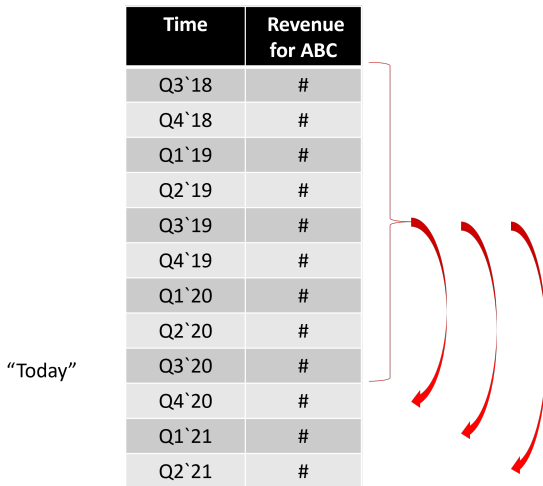
If the model is stable, these should yield similar results.

Multi Step Ahead Forecasts - Static

Static Multi Step Ahead forecasts use the same set of information to forecast multiple horizons

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Multi Step Ahead Forecasts - Dynamic

Dynamic forecasts use the one step ahead estimates to forecast two steps ahead, etc..

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"Today"

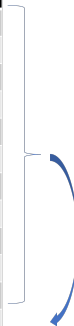
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Q4`20	2
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Q2`21	



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Q3`20	#
Q4`20	2
Q1`21	2.25
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Multi Step Ahead Forecasts - Dynamic

Dynamic forecasts use the one step ahead estimates to forecast two steps ahead, etc..

"Today"

Time	Revenue for ABC
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Q4`18	#
Q1`19	#
Q2`19	#
Q3`19	#
Q4`19	#
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- Advantages: Can potentially extend far into the future; adapt quickly to new settings

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Q: What are (dis)advantages of Dynamic relative to Static?

A:

- Advantages: Can potentially extend far into the future; adapt quickly to new settings
- Disadvantage: Errors compound on each other

Mean and Linear Time Trends

Mean and Linear Time Trends

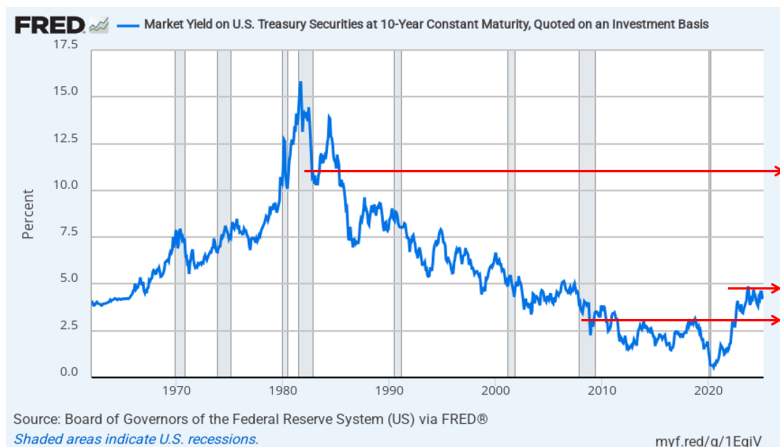
The two “simplest” models can serve as naive benchmarks:

$$\text{Mean Model: } E_t[Y_{t+h}] = \frac{1}{T} \sum_{t=1}^T Y_t$$

$$\text{Linear Time Trend Model: } E_t[Y_{t+h}] = \hat{\alpha} + \hat{\beta} \times \text{Time}_{t+h}$$

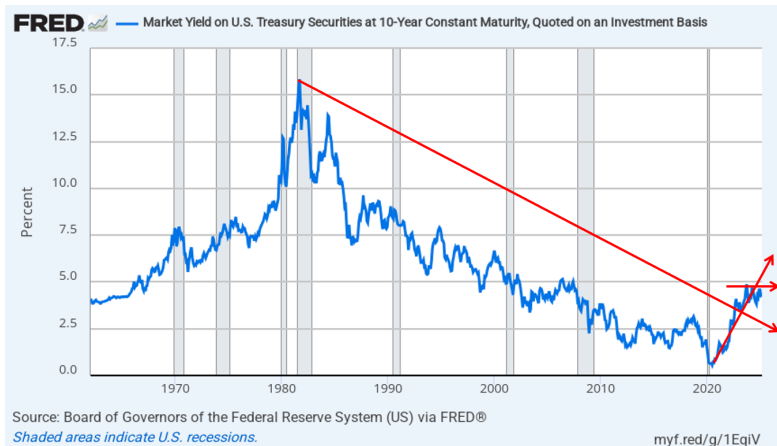
Mean Model

Though easy to compute, the size of training sample must be tuned



Linear Time Trend Model

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Training Sample Selection

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A few best practices:

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- Ideally, Training Sample would “look like” the current period
- Ideally, Testing Sample would “look like” the out of sample period
- Try several sample splits and examine the range of forecasts for stability

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

5 Forecast

- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
 - MA
 - AR
 - ARMA
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models

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- Imagery: every series has its own unique temporal “fingerprint” defined by the mean, variance, and autocorrelation of the series.
- If we can identify the fingerprint, we can model the series.

Moving Average Process

$$Y_t = \mu + \varepsilon_t + \alpha\varepsilon_{t-1} \quad (16)$$

where $\varepsilon_t \sim N(0, \sigma^2)$ iid.

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Example (MA(1))

- Find the $E[Y_t]$.
- Find the $Var(Y_t)$.
- Find the autocovariance of Y over 1 lag i.e. $Cov(Y_t, Y_{t-1})$.
- Find the autocovariance of Y over 2 lags i.e. $Cov(Y_t, Y_{t-2})$.

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Moving Average Process

Definition

A moving average process of order q [MA(q)] can be represented by

$$Y_t = \mu + \varepsilon_t + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}$$

where $\varepsilon \sim N(0, \sigma^2)$ iid

Autoregressive Process

Consider the $MA(\infty)$

$$Y_t = \mu + \sum_{j=0}^{\infty} \alpha_j \varepsilon_{t-j}$$

where $\alpha_0 \equiv 1$, and $\varepsilon \sim N(0, \sigma^2)$ iid. Assume that the α weights converge to zero as j gets large.

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$$Y_t = \mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j}$$

Autoregressive Process

Notice from the last line above that

$$Y_{t-1} = \mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j-1}$$

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and so we can write

$$Y_t - \theta Y_{t-1} = \mu - \theta\mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j} - \theta \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j-1}$$

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where $\delta = \mu - \theta\mu$.

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where $\delta = \mu - \theta\mu$. Defining $y_t = Y_t - \mu$ yields an AR(1) $y_t = \theta y_{t-1} + \varepsilon_t$

Example (AR(1))

Consider the AR(1) $y_t = \theta y_{t-1} + \varepsilon_t$, where $\varepsilon \sim N(0, \sigma^2)$ iid.

- Find the $E[y_t]$.
- Find the $\text{Var}(y_t)$.
- Find the autocovariance for Y_t and Y_{t-1} .
- Find the autocovariance for Y_t and Y_{t-2} .

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An autoregressive process of order p [AR(p)] can be represented by

$$y_t = \sum_{j=1}^p \theta_j y_{t-j} + \varepsilon_t$$

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Definition

An ARMA(p,q) process can be represented by

$$y_t = \sum_{j=1}^p \theta_j y_{t-j} + \varepsilon_t + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}$$

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In practice, $y_0, y_{-1}, \dots$ are not observed. So, we cut off the infinite sum and optimize

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Estimation

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$S(\alpha) = \sum_{t=2}^T (y_t - \alpha \sum_{j=0}^{\infty} (-\alpha)^j y_{t-j-1})^2$.

In practice, $y_0, y_{-1}, \dots$ are not observed. So, we cut off the infinite sum and optimize

$$S(\alpha) = \sum_{t=2}^T (y_t - \alpha \sum_{j=0}^{t-2} (-\alpha)^j y_{t-j-1})^2$$

Notice this is non linear.

Estimation

Typical OLS estimation is difficult in MA or ARMA models.

Consider an MA(1): $y_t = \varepsilon_t + \alpha\varepsilon_{t-1}$.

OLS would minimize $S(\alpha) = \sum_{t=2}^T (y_t - \alpha\varepsilon_{t-1})^2$.

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Notice this is non linear. Maximum Likelihood is the common alternative.

Model Choice via Box Jenkins

Rules of Thumb

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An AR(p) process is described by

- ① An ACF with slow decay
- ② A PACF that is (close to) zero for lags larger than p

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An ARMA process is described by

- Slow decay in both ACF and PACF

Example (Reading the Correlogram)

Q: What model specification seems appropriate given the correlogram below?

Date: 07/11/07 Time: 11:27

Sample: 1996M04 2007M06

Included observations: 135

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	0.680	0.680	63.865	0.000
		2	0.549	0.160	105.71	0.000
		3	0.495	0.139	140.06	0.000
		4	0.466	0.108	170.72	0.000
		5	0.402	0.001	193.72	0.000
		6	0.331	-0.031	209.45	0.000
		7	0.305	0.034	222.91	0.000
		8	0.271	-0.003	233.64	0.000
		9	0.248	0.021	242.68	0.000
		10	0.239	0.042	251.10	0.000
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A: The slowly decaying ACF and the spike at 1 lag for the PACF suggest an AR(1) might be appropriate.

Diagnostic Checking

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- ④ Search for the smallest AIC and/or BIC values
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ARIMA is an extension of the ARMA structure,

AR: Autoregressive

I: Integrated

MA: Moving Average

The order of integration indicates the number of times the series must be differenced in order to achieve stationarity. A process that is $I(0)$ is stationary. A process that is $I(1)$ is said to be “integrated of order 1”, implying that it must be differenced once to achieve stationarity.

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Extension: Seasonal ARIMA

- $\text{SARIMA}(p, d, q) \times (P, D, Q)_s$
- Captures both short-run dynamics and seasonal patterns
- Seasonal differencing: $\Delta_s Y_t = Y_t - Y_{t-s}$
- Connection: Seasonal ARIMA underlies many decomposition tools (e.g. X-13 ARIMA-SEATS)

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

5 Forecast

- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models

● ARIMAX

● VAR

- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating

ARIMAX Motivation: Diagnosing Missing Structure

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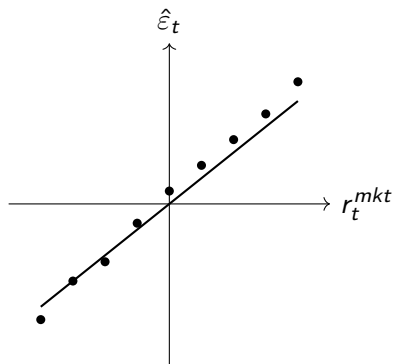
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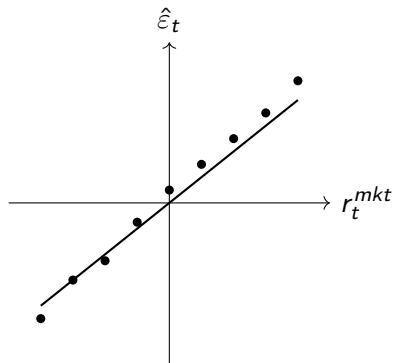


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- If residuals contain structure
 $\Rightarrow$ model is misspecified
- Add X_t to capture this variation
- $\Rightarrow$ ARIMAX

ARIMAX Model

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- Examples of X_t :
 - Market returns
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- Interpretation:
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- Important:
 - Must forecast X_{t+h} or use only lagged X
 - If forecasting X_{t+h} , then forecast errors may compound

Case Study Background

Consider the Phillips Curve, which relates inflation (π) to unemployment (UR). Suppose each follows an AR(1), and by theory is related to the other.

$$\begin{aligned}\pi_t &= \delta_1 + \theta_{11}\pi_{t-1} + \theta_{12}UR_{t-1} + \epsilon_{1t} \\ UR_t &= \delta_2 + \theta_{21}\pi_{t-1} + \theta_{22}UR_{t-1} + \epsilon_{2t}\end{aligned}$$

where ϵ_{1t} and ϵ_{2t} are potentially correlated white noise processes, each independent of the history of UR and π .

The Case for VAR

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- Possibly lead to more accurate forecasts.
- Greene (2000) suggests that simultaneous estimates might be more efficient in some situations.

A VAR(1) System

The VAR(1) specification can be written as

$$\begin{pmatrix} Y_t \\ X_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ X_{t-1} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

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or written compactly as

$$\vec{Y}_t = \delta + \Phi_1 \vec{Y}_{t-1} + \vec{\epsilon}_t$$

where $\vec{Y}_t = (Y_t, X_t)'$ and $\vec{\epsilon}_t = (\epsilon_{1t}, \epsilon_{2t})'$.

A VAR(p) System

Define $\vec{Y}_t$ as a $t \times k$ matrix. Then the VAR(p) model can be written as

$$\vec{Y}_t = \delta + \Phi_1 \vec{Y}_{t-1} + \cdots + \Phi_p \vec{Y}_{t-p} + \vec{\epsilon}_t$$

where each parameter matrix Φ is a $k \times k$ and the white noise errors $\vec{\epsilon}$ is a k -dimensional vector with covariance matrix Σ .

The expected value can be written as

$$E\{\vec{Y}_t\} = \delta + \Phi_1 E\{\vec{Y}_{t-1}\} + \cdots + \Phi_p E\{\vec{Y}_{t-p}\}$$

If we define the lag operator L and I_k as the k -dimensional Identity matrix, then we can rewrite as

$$\mu = E\{\vec{Y}_t\} = (I_k - \Phi_1 - \cdots - \Phi_p)^{-1} \delta = \Phi(1)^{-1} \delta$$

Note that stationarity requires that $\Phi(1)$ is invertible.

Covariance Estimation

Denote $e_{1t}, \dots, e_{kt}$ as the residuals from each of the k -equations in the system. Then we can estimate the (i,j) -element of Σ as

$$\hat{\sigma}_{ij} = \frac{1}{T-p} \sum_{t=p+1}^T e_{it} e_{jt}$$

This permits us to form

$$\hat{\Sigma} = \frac{1}{T-p} \sum_{t=p+1}^T \vec{e}_t \vec{e}_t'$$

where $\vec{e}_t = (e_{1t}, \dots, e_{kt})$.

Forecasting 1-step ahead

If we demean via $\vec{y}_t = \vec{Y}_t - \mu$, we can form

$$\vec{y}_t = \Phi_1 \vec{y}_{t-1} + \cdots + \Phi_p \vec{y}_{t-p} + \vec{\epsilon}_t$$

then we can form optimal one-period ahead forecasts via

$$\vec{y}_{T+1|T} = E\{\vec{y}_{T+1} | \vec{y}_T, \vec{y}_{T-1}, \dots\} = \Phi_1 \vec{y}_T + \cdots + \Phi_p \vec{y}_{T-p+1}$$

with the accompanying forecast error variance

$$V\{\vec{y}_{T+1} | \vec{y}_T, \vec{y}_{T-1}, \dots\} = \Sigma$$

Forecasting multiple steps ahead

Forecasting is accomplished via recursive substitution. For instance, for a

$$\begin{aligned}\vec{y}_{T+2|T} &= \Phi_1 \vec{y}_{T+1|T} + \cdots + \Phi_p \vec{y}_{T-p+2|T} \\ &= \Phi_1 (\Phi_1 \vec{y}_T + \cdots + \Phi_p \vec{y}_{T-p+1|T}) + \cdots + \Phi_p \vec{y}_{T-p+2|T}\end{aligned}$$

Determining the Lag Length (p)

Determining the lag length (p) is not easy. We cannot use autocorrelograms (see Canova 1995).

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A reasonable strategy (see Verbeek 2000) is to

- 1 Estimate the VAR over a range of p values
- 2 Pick the specification with the smallest AIC/BIC.

Impulse Response Function (IRF)

If $\Phi(1)$ is invertible, we can rewrite the VAR as a Vector Moving Average (VMA)

$$\vec{Y}_t = \mu + \Phi(L)^{-1} \vec{\epsilon}_t$$

Writing $\Phi(L)^{-1} = I_k + A_1L + A_2L^2 + \dots$, we can see

$$\vec{Y}_t = \mu + \vec{\epsilon}_t + A_1 \vec{\epsilon}_{t-1} + A_2 \vec{\epsilon}_{t-2} + \dots$$

Each element of any A matrix can be thought of as a marginal effect of a shock to the residual upon data in Y .

Impulse Response Function (IRF)

Consider again the Phillips Curve example

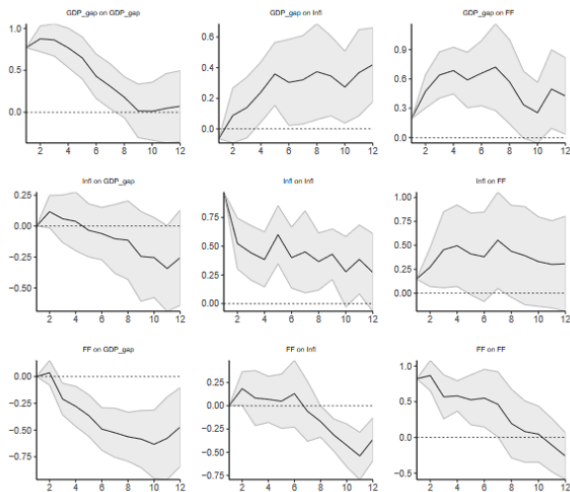
$$\begin{pmatrix} UR_t \\ \pi_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} UR_{t-1} \\ \pi_{t-1} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

Rewriting as a VMA we have $\vec{Y}_t = \mu + \vec{\epsilon}_t + A_1 \vec{\epsilon}_{t-1} + A_2 \vec{\epsilon}_{t-2} + \dots$

To examine the impact of a unit-one shock of unemployment upon inflation we consider the dynamics of the marginal effects

- $\frac{\partial Y_{2,t+1}}{\partial \epsilon_{1,t}} = A_{1,(1,2)}$
- $\frac{\partial Y_{2,t+2}}{\partial \epsilon_{1,t}} = A_{2,(1,2)}$
- ...

Impulse Response Function (IRF)



Multivariate Cointegration

Consider a binary VAR(1) model with variables integrated of order one ($\vec{Y}_t \sim I(1)$), we can write it as:

$$\vec{Y}_t = \delta + \Phi_1 \vec{Y}_{t-1} + \vec{\epsilon}_t$$

where matrix Φ_1 is a 2×2 and the white noise errors $\vec{\epsilon}$ is a 2-dimensional vector with covariance matrix Σ .

By subtracting the lagged vectors $\vec{Y}_{t-1}$ from both side, we get:

$$\begin{aligned}\vec{Y}_t - \vec{Y}_{t-1} &= \delta + \Phi_1 \vec{Y}_{t-1} - \vec{Y}_{t-1} + \epsilon_t \\ \Delta \vec{Y}_t &= \delta + (\Phi_1 - I_2) \vec{Y}_{t-1} + \epsilon_t\end{aligned}$$

Multivariate Cointegration

- Since we know Y_t is integrated of order one, so we have an $I(0)$ on the left hand side.
- The right hand side is the combination of two $I(0)$ (one constant vector δ and white noise vector $\vec{\epsilon}$) and $(\Phi_1 - I_2)\vec{Y}_{t-1}$.
- If all variables are not cointegrated, then we must have:

$$(\Phi_1 - I_2) = \vec{0} \Rightarrow \Phi_1 = I_2$$

- If the variables are cointegrated, then there exists $r(0 < r \leq 2)$ cointegrated relations, $(\Phi_1 - I_2)\vec{Y}_{t-1}$ can be written as a $I(0)$ variable:

$$(\Phi_1 - I_2)\vec{Y}_{t-1} = \lambda\gamma'\vec{Y}_{t-1}$$

where γ' is the $(r \times 2)$ matrix of cointegration coefficients and λ is a $(2 \times r)$ matrix. Specifically, if $r = 2$, then all variables in $\vec{Y}_t$ must be stationary.

Multivariate Cointegration

The above structure can be extended to VAR(p) model with k variables with integrated of order d ($\vec{Y}_t \sim I(d)$):

$$\vec{Y}_t = \delta + \Phi_1 \vec{Y}_{t-1} + \cdots + \Phi_p \vec{Y}_{t-p} + \vec{\epsilon}_t$$

where each parameter matrix Φ is a $k \times k$ and the white noise errors $\vec{\epsilon}$ is a k -dimensional vector with covariance matrix Σ .

Apply difference operator $\Delta = 1 - L$ to both side, we get:

$$\begin{aligned}\Delta \vec{Y}_t &= \delta + \left(\sum_{j=1}^p \Phi_j - I_k \right) \vec{Y}_{t-1} + \Gamma_1 \Delta \vec{Y}_{t-1} + \cdots + \Delta \Gamma_p \vec{Y}_{t-p-1} + \vec{\epsilon}_t \\ &= \delta + \Pi \vec{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \vec{Y}_{t-j} + \vec{\epsilon}_t\end{aligned}$$

This equation also called vector error correction model (VECM).

Multivariate Cointegration

$$\Delta \vec{Y}_t = \delta + \Pi \vec{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \vec{Y}_{t-j} + \vec{\epsilon}_t$$

The rank of matrix Π determines number of cointegration vectors:

- If $\text{rank}(\Pi) = 0$, all variables in vector $\vec{Y}_t$ are non-stationary. There is no linear combination of variables that leads to stationary
- If Π has full rank k , then all variables in $\vec{Y}_t$ must be stationary
- If Π has reduce rank r ($0 < r < k$), then there are cointegration relations among $\vec{Y}_t$

Vector Error Correction Model

Consider VECM equation:

$$\Delta \vec{Y}_t = \delta + \Pi \vec{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \vec{Y}_{t-j} + \vec{\epsilon}_t$$

If Π has rank $r > 0$, then Π can be written as production of A and B' :

$$\Delta \vec{Y}_t = \delta + AB' \vec{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \vec{Y}_{t-j} + \vec{\epsilon}_t$$

where A is a $(k \times r)$ matrix contains the cointegration vectors and B is a $(k \times r)$ matrix which contains the speed of adjustment parameters.

We call this new version VECM as restricted regression in the sense that it imposes the hypothesis that $\text{rank}(\Pi) > 0$, so the original VECM is unrestricted regression.

Johansen Test

As mentioned before, the Engle and Granger two-step test cannot be used to test cointegration involves more than two variables, so here we use Johansen test for multivariate cointegration testing.

Johansen test involves trace test and maximum eigenvalue test:

- Trace test:

$$\lambda_{trace} = -T \sum_{i=r+1}^k \log(1 - \lambda_i)$$

$$H_0 : r < k \iff H_1 : r = k$$

- Maximum eigenvalue test:

$$\lambda_{max} = -T \log(1 - \lambda_{r+1})$$

$$H_0 : r = q, q = 1, 2, \dots, k \iff H_1 : r \leq q + 1$$

where λ_i is the eigenvalue value of matrix Π .

In most cases, it is hard to tell the causality among variables from regression. For example:

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

the significance of coefficient β_1 indicates that there exists relationship between variable y and X , but says nothing about X cause y .

For causality, we have the following concepts:

- If X causes y , the changes of X happened before and were followed by the changes of y
- If X causes y , including X in regression helps improve forecasting of y , while reverse is wrong

Granger(1969) and Sims(1972) use VAR to test for the “Causality” in above sense, thus this causal relationship is called *Granger*(or *Granger-Sims*) *causality*.

Granger Causality

Consider a binary (y and X) system to keep things simple. Based on the “Causality” concepts:

- X is said to Granger-cause y if the current or lagged term of X help improve forecast X
- X fails to Granger-cause y if predict accuracy of y_{t+s} ($s > 0$) based on information $(y_t, y_{t-1}, \dots)$ is the same as predict based on information $(y_t, y_{t-1}, \dots, X_t, X_{t-1}, \dots)$
- If we restrict the relationship between X and y are linear, we may use MSE to measure forecasting performance and the above condition can be rewrite as: X fails to Granger-cause y if

$$MSE[\hat{E}(y_{t+s}|y_t, y_{t-1}, \dots)] = MSE[\hat{E}(y_{t+s}|y_t, y_{t-1}, \dots, X_t, X_{t-1}, \dots)]$$

Example (VAR Formulation)

The binary VAR(p) specification can be written as

$$\begin{pmatrix} Y_t \\ X_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \Phi_{11}^1 & \Phi_{12}^1 \\ \Phi_{21}^1 & \Phi_{22}^1 \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ X_{t-1} \end{pmatrix} + \dots + \begin{pmatrix} \Phi_{11}^p & \Phi_{12}^p \\ \Phi_{21}^p & \Phi_{22}^p \end{pmatrix} \begin{pmatrix} Y_{t-p} \\ X_{t-p} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

If X fails to Granger-cause y , then we have:

$$\Phi_{12}^1 = \Phi_{12}^2 = \dots = \Phi_{12}^p = 0$$

that is this regression has a lower triangle coefficient matrix.

Granger Causality

Consider a linear regression of X_t on the past, current and future value of y 's:

$$X_t = c + \sum_{j=0}^{\infty} b_j y_{t-j} + \sum_{j=0}^{\infty} d_j y_{t+j} + \epsilon_t$$

where the exogenous assumption is satisfied ($E[\epsilon_t y_s] = 0$ for all t and s), then X fails to Granger-cause y if and only if $d_j = 0$ for $j = 1, 2, \dots$

Similarity, we can use this regression to test whether y Granger-cause X :

$$y_t = c + \sum_{j=0}^{\infty} b_j X_{t-j} + \sum_{j=0}^{\infty} d_j X_{t+j} + v_t$$

where the exogenous assumption is satisfied ($E[v_t X_s] = 0$ for all t and s), then y fails to Granger-cause X if and only if $d_j = 0$ for $j = 1, 2, \dots$

Granger Causality

Causality model:

$$y_t = c_0 + \sum_{i=0}^{p_y} \alpha_i y_{t-i} + \sum_{j=0}^{p_x} \beta_j x_{t-j} + u_t$$

where p_x and p_y are the order of AR model of each series which are determined by F-test and/or AIC, BIC.

Granger-Causality test:

- Test: H_0 (X fail to Granger-cause y): $\beta_1 = \dots = \beta_{p_x} = 0$
- Estimate the unrestricted regression, keep RSS_U
- Estimate the restricted regression ($\beta_1 = \dots = \beta_{p_x} = 0$), keep RSS_R
- Construct F -test: $F = [(T - k)/q] \times [(RSS_R - RSS_U)/RSS_U]$,
where $k = (1 + p_x + p_y)$ and $q = (1 + p_x)$

(Similar test can be conducted to test whether y Granger-cause X)

There are four possible outcome for Granger-Causality test:

- X Granger-cause y , but y fails to Granger-cause X
- y Granger-cause X , but X fails to Granger-cause y
- X Granger-cause y and y Granger-cause X
- X fails to Granger-cause y and y fails to Granger-cause X

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- Tradeoff:
 - Flexibility vs interpretability

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- Use time series EDA to select lag structure

Date	Y	X	YL1	YL2	XL1	XL2	XL3
Jan-00	10	100					
Feb-00	11	112	11		100		
Mar-00	12	115	12	10	112	100	
Apr-00	13	118	13	11	115	112	100
May-00	14	120	14	12	118	115	112
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- Once we prepare the data in this way, most of our ML models can be implemented without major changes.

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- Trees capture:
 - Nonlinearities
 - Threshold effects
 - Interactions

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- Time series caution:
 - Use block bootstrap to preserve temporal dependence

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- Examples:
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- Estimation:
 - Minimize forecast error via gradient descent
 - Learning rate η : controls step size of updates

From Lagged Inputs to Sequential Models

- Neural networks for time series use lagged inputs:

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- Interpretation:
 - h_t is a learned summary of the past

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 - LSTM Learns what history matters for forecasting

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- Implications for forecasting:
 - Captures long-range dependencies
 - More stable learning over long horizons

AR vs Neural Network vs LSTM

- All models estimate:

$$f_h(x_t) = E[Y_{t+h} \mid x_t]$$

Model	Input	Memory	Nonlinearity
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- AR: hand-specified memory
- NN: nonlinear but still fixed memory
- LSTM: learns what to remember

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- NN: nonlinear transformation of fixed lags
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From fixed memory $\rightarrow$ learned memory

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- Alternative:
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 - Imposes the same parameter ϕ for all i
- Nonlinear relationships
 - Model is restricted to linear form
 - Structural limitation $Y_t^{(i)} = \phi Y_{t-1}^{(i)}$

- Replace linear model with:

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- Addresses key limitations:
 - Nonlinearity: $f_h(\cdot)$ is flexible
 - Heterogeneity: response varies with the input (state-dependent dynamics)

Training a Global Model

- Training data consists of:

$$\{(x_t^{(i)}, Y_{t+h}^{(i)})\}_{i,t}$$

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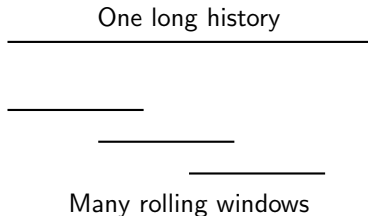
	Y^a	Y^b	Y^c	X^a	X^b	X^c	
t	.	.	.	.	.	.	Panel 1
$t+1$	.	.	.	.	.	.	Panel 2
$t+2$	.	.	.	.	.	.	Panel 3
$t+3$	.	.	.	.	.	.	
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Time Depth vs Cross-Sectional Learning

- Suppose we have many time series over long horizons

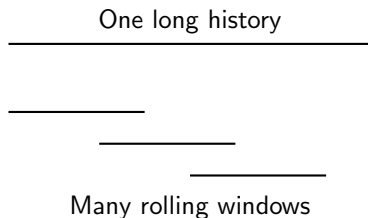
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- Two approaches:
 - One long history per series
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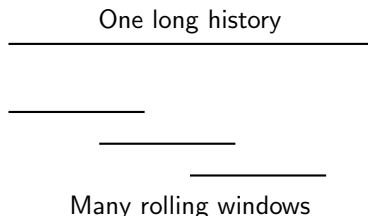
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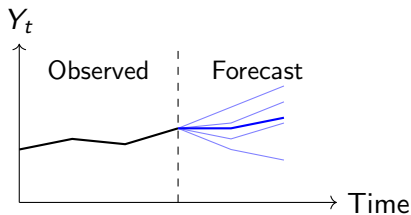
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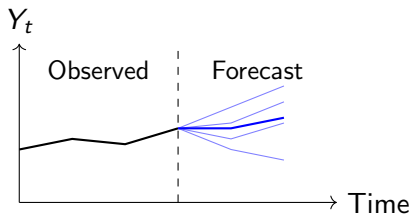
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- Forecast generated via simulated paths
- Captures uncertainty over future outcomes

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- Summary:
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- **Problem:** Forecasts do not respect aggregation constraints

Global models improve forecasts, but do not enforce consistency

Hierarchical Forecasting: The Solution

- Goal: Ensure forecasts are **coherent**

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- General form:

$$\tilde{y} = S\hat{y}$$

- S : aggregation structure
- $\hat{y}$: base forecasts
- $\tilde{y}$: coherent forecasts

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- **Middle-out:**

- Forecast at an intermediate level, then aggregate/distribute

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- **Step 2: Distribute down within each region**
 - Region 1 (historical shares: 60% / 40%): $\tilde{Y}_A = 42, \quad \tilde{Y}_B = 28$
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- **Key idea:**
 - Forecast where signal-to-noise is best (middle level)
 - Aggregate up, allocate down

Key Insight: Model vs Structure

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- Model-agnostic:

- ARIMA
- Machine Learning
- DeepAR / Global Models
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Hierarchical methods sit on top of any forecasting model

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- This leads to **foundation models**

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- Interpretation:
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- Extends:
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 - RNN / LSTM models

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- Key distinction:
 - Training from scratch: initialize θ randomly
 - Fine-tuning: initialize at $\theta_{\text{pretrained}}$
- Compared to initial estimation, fine tuning
 - Usually has smaller learning rates
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Fine-Tuning Foundation Models

- Start from pretrained parameters:

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- Update parameters using new data:

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- Interpretation:
 - Adjust a general model to a specific dataset

Global vs Foundation Models

	Global Models	Foundation Models
Training data	Single panel	Many datasets
Training	From scratch	Pretrained once
Parameters	Task-specific	Reused across tasks
Setup	Moderate	Minimal

Global vs Foundation Models

	Global Models	Foundation Models
Training data	Single panel	Many datasets
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- Foundation models extend global modeling to large-scale learning

When to Use Foundation Models

- Useful when:
 - Limited data
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- Summary:
 - Trade customization for scalability

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$$[L_{t+h}, U_{t+h}]$$

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Simulation-Based Uncertainty

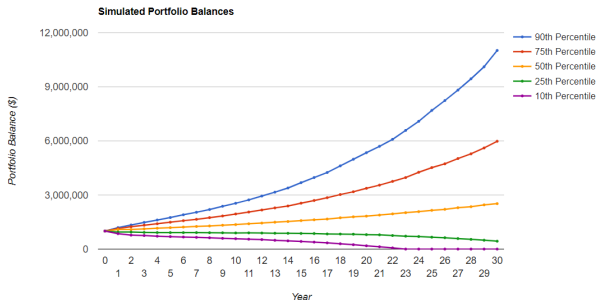
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- When simulating random noise, be sure to set your seed for the random number generator to ensure you can replicate results.

Simulating a Random Walk

Definition (Geometric Random Walk)

Denote P_t as the price at time t , the simple return as $R_t = \frac{P_t}{P_{t-1}} - 1$, and the log return as $r_t = \ln(P_t/P_{t-1})$. Notice that $\frac{P_t}{P_{t-H}} = 1 + R_t(H) = \exp(r_t + \dots + r_{t-H+1})$, which implies

$$P_t = P_0 \exp(r_t + \dots + r_1)$$

If $r_1, r_2, \dots$ are iid $N(\mu, \sigma^2)$ then we call the price process a lognormal geometric random walk with parameters (μ, σ^2) .

Monte Carlo - Why GEOMETRIC Random Walk?

Why not just use a simple random walk such that $P_t = P_{t-1} + R_t$, where $R_t \sim N(\mu, \sigma^2)$ iid is the simple return iid?

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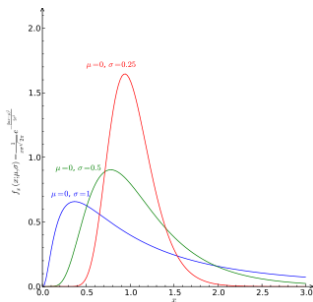
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Solution: If Gross Returns $(1 + R)$ are log-Normal, then log returns $r \equiv (1 + R)$ are Normal

Definition (Log Normal Distribution)

$$f(X) = \frac{1}{X(2\pi)^{1/2}\sigma} \exp \left[-\frac{1}{2} \left(\frac{\ln(X) - \mu}{\sigma} \right)^2 \right]$$

for $0 < X < \infty$. If $\ln(X) \sim N$, then $X \sim \text{log Normal}$. Especially useful for prices, rather than returns.



Monte Carlo - Why GEOMETRIC Random Walk?

Resolving Limited Liability

Recall $r_t = \ln(1 + R_t)$. Notice that $\exp(r_t) = \exp(\ln(1 + R_t)) = 1 + R_t$. Recall that the exponential function has positive range, implying that $(1 + R_t) \geq 0 \rightarrow R_t \geq -1 \rightarrow P_t \geq 0$.

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Resolving the lack of Multi-Period Gaussianity

$$\begin{aligned} 1 + R_t(H) &= (1 + R_t)(1 + R_{t-1})\dots \\ &= \exp(r_t)\exp(r_{t-1})\dots \\ &= \exp(r_t + r_{t+1} + \dots) \\ \ln(1 + R_t) &= r_t + r_{t-1} + \dots \end{aligned}$$

Recall that the **sum** of independent Gaussian r.v.'s is Gaussian.

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- Useful when:
 - Distribution is unknown
 - Model assumptions are unreliable

Original Data		Bootstrap Sample 1	Bootstrap Sample 2	Bootstrap Sample 3
1999		2001	2005	2002
2000		2006	1999	2006
2001		2000	2001	2003
2002		1999	2003	2001
2003		2004	2006	1999
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Variants:

- Circular bootstrap
- Stationary bootstrap

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- 4 Compute the bootstrapped standard error $se(\hat{\theta}) = \sqrt{\frac{1}{B-1} \sum (\hat{\theta}_b - \bar{\hat{\theta}})^2}$

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Goal: Forecast a fixed future outcome as information evolves

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Same target, different information sets

Naïve Approach: Separate Forecasting Problems

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Limitation:

- Forecast updates are not linked
- Each revision comes from a new model

We re-learn the model each time

Reframing the Problem

Rather than separate models We have one evolving system

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New idea:

- Do not re-estimate from scratch
- Update our **belief about the system**

Forecast updates should come from updated beliefs, not new models

State Space Models

Represent the system using latent states:

State equation:

$$x_t = Fx_{t-1} + \eta_t$$

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Key idea:

- The system evolves through x_t
- We learn about it through Y_t

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Forecast depends on:

- Current estimate of the latent state

Updating the Forecast: Kalman Filter

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Key result:

- Update state $\rightarrow$ update forecast
- No re-estimation of model parameters

Kalman Gain: How Much Do We Update?

Update step: $\hat{x}_t = \hat{x}_{t|t-1} + K_t \underbrace{(Y_t - H\hat{x}_{t|t-1})}_{\text{forecast error}}$

Kalman Gain: $K_t = P_{t|t-1} H' (H P_{t|t-1} H' + R)^{-1}$

Interpretation:

- K_t controls how much we adjust our belief using new data
- Balances:
 - $P_{t|t-1}$: uncertainty in the model (state)
 - R : noise in the data (measurement error)

Intuition:

- Large R (noisy data) $\Rightarrow$ small $K_t \rightarrow$ trust the model
- Large $P_{t|t-1}$ (uncertain model) $\Rightarrow$ large $K_t \rightarrow$ trust the data

Kalman Gain = optimal weight on new information

Interpretation: Forecast Updating

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Task: Forecast Volatility

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2015-02-05: **2,062.52** Index (+ see more)

Daily, Not Seasonally Adjusted, SP500, Updated: 2015-02-06 7:00 AM CST

Click and drag in the plot area or select dates: 1yr | 5yr | 10yr | Max

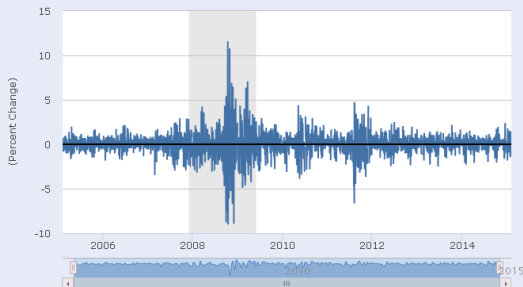
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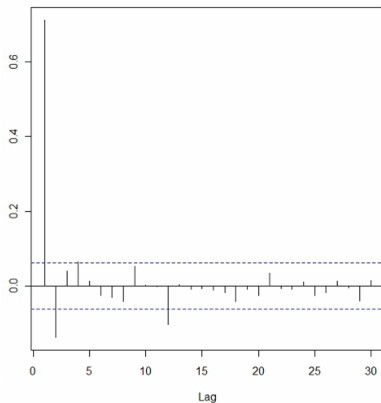


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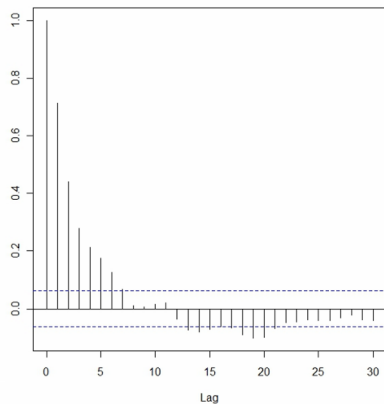
Shaded areas indicate US recessions - 2015 research.stlouisfed.org

Autocorrelations and Volatility

ACF (Returns)



ACF (Returns Squared)



A Few Stylized Facts

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- Substantial comovement in volatility across assets, markets, and countries

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- $\frac{(\ln H_t - \ln L_t)^2 t}{4 \ln 2}$, where $H(L)$ is the High(Low) for the day
- $0.5(\ln \frac{H_t}{L_t})^2 - .39(\ln \frac{C_t}{O_t})^2$, where $C(O)$ is the Close(Open) for the day

$$\begin{aligned}r_{t+j\Delta} &= p_{t+j/M} - p_{t+(j-1)/M} \\RV_{t+1}(\Delta) &\equiv \sum_{j=1}^M r_{t+j\Delta}^2\end{aligned}$$

where p is the log price, M refers to the number of intraday equally spaced returns over the trading day t , and $\Delta = 1/M$ denotes the sampling interval.

Example: If we sample every half hour, then $M = 13$.

GARCH(p,q)

$$\begin{aligned}r_t &= \mu + \varepsilon_t & \varepsilon_t &\sim N(0, h_t) \\ \varepsilon_t &= z_t \sqrt{h_t} & z_t &\sim N(0, 1) \\ h_t &= \omega + \sum_{j=1}^p \beta_j h_{t-j} + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2\end{aligned}$$

where $\sum_{j=1}^p \beta_j h_{t-j}$ is the GARCH term and $\sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$ is the ARCH term. All parameters are positive sum of ARCH and GARCH terms < 1 .

Workhorse model:

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Workhorse model: GARCH(1,1) $h_t = \omega + \beta h_{t-1} + \alpha \varepsilon_{t-1}^2$

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- $E[\varepsilon_t] = 0$
- $Var[\varepsilon_t] = E[z_t^2] \times E[\omega + \alpha_1 \varepsilon_{t-1}^2] = \frac{\omega^2}{1-\alpha_1}$
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The fact that the conditional variance depends upon its own past drives the autoregressive conditional heteroscedasticity.

Testing for ARCH Effects

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Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

5 Forecast

- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

Forecasting: Model Selection Map

Data / Structure	Recommended Approach
Single series, short history	ARIMA, ETS
Single series + exogenous drivers	ADL, ARIMAX, ECM
Multiple interacting series	VAR (VECM if cointegrated)
Many related series (panel)	Global models (DeepAR, panel regression)
Nonlinear patterns / complex features	Machine Learning (RF, XGBoost)
Long sequences / rich temporal patterns	Deep Learning (LSTM, N-BEATS)